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Analysis of a Stochastic HBV Infection Model with DNA-Containing Capsids and Virions

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Abstract
In this paper, we propose a stochastic differential equation model which is used to explore how the environmental noise affects the spread of hepatitis B virus. Firstly, we show that there exists a unique global positive solution of the stochastic system with any positive initial value. Then, we adopt a stochastic Lyapunov function method to establish sufficient conditions for the existence and uniqueness of an ergodic stationary distribution of the proposed stochastic model. Especially, under the same conditions as the existence of a stationary distribution, it is worth noting that we obtain the specific form of probability density around the quasi-infected steady state of the stochastic system. Thirdly, we obtain sufficient criteria for extinction of the infected hepatocytes. Finally, numerical simulations are introduced to validate the theoretical findings.

Keywords Hepatitis B · Stationary distribution · Ergodicity · Fokker–Planck equation · Probability density · Extinction

Mathematics Subject Classification 34E10 · 60H10 · 92B05 · 92D30

1 Introduction

Hepatitis B is a dangerous infectious disease caused by the hepatitis B virus (HBV) (Yang et al. 2022). It remains a serious public health problem which attacks and infects the healthy liver cells. According to the World Health Organization reports, an estimated two billion people infected by HBV, and over 320 million having chronic liver infections (WHO https://www.who.int/health-topics/hepatitistab=tab_

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1; Ott et al. 2012). That is why it is important to understand the transmission mechanism of HBV.

Mathematical models have been frequently used as tools to understand the infection and pathogenesis mechanism of some infectious diseases and some chronic viral infections. They can provide us some new ideas for the prevention and control of the diseases (Yang et al. 2022; Danane et al. 2020; Wang et al. 2022; Manna and Chakrabarty 2015; Gourley et al. 2008; Li and Zu 2019; Huang et al. 2010). Recently, many different mathematical models have been developed since the pioneer work of Nowak and Bangham (1996). Some models studied the dynamics of HBV infection with antibody immune response (Danane et al. 2020). Some models considered the time from infection to the release of free virus particles (Gourley et al. 2008; Huang et al. 2010). In particular, Manna and Chakrabarty (2015) proposed a HBV model which takes into account both uninfected and infected hepatocytes along with the intracellular HBV DNA-containing capsids:

$$\begin{cases} \dfrac{\mathrm{d}H(t)}{\mathrm{d}t} = s - kH(t)V(t) - \mu H(t), \\ \dfrac{\mathrm{d}I(t)}{\mathrm{d}t} = kH(t)V(t) - \delta I(t), \\ \dfrac{\mathrm{d}D(t)}{\mathrm{d}t} = aI(t) - \beta D(t) - \alpha D(t), \\ \dfrac{\mathrm{d}V(t)}{\mathrm{d}t} = \beta D(t) - cV(t), \end{cases} \tag{1.1}$$

where $H(t)$ and $I(t)$ represent the numbers of the uninfected and the infected hepatocytes at time t, respectively. $D(t)$ and $V(t)$ denote the numbers of the intracellular HBV DNA-containing capsids and the virions in plasma at time t, respectively. All parameters are strictly positive and their descriptions are given in Table 1. From Manna and Chakrabarty (2015), the authors obtained the basic reproduction number R_0 for system (1.1) as follows:

$$R_0 = \frac{a\beta sk}{c\delta\mu(\alpha+\beta)}.$$

And if $R_0 \leq 1$, the uninfected steady state $E_0 = (H_0, 0, 0, 0) = (s/\mu, 0, 0, 0)$ is globally asymptotically stable in the positive invariant set Ω, where $\Omega := \{(H, I, D, V) \in \mathbb{R}_+^4 : 0 \leq H, I \leq s/\mu, 0 \leq D \leq as/(\mu(\alpha+\beta)), 0 \leq V \leq a\beta s/(c\mu(\alpha+\beta))\}$. If $R_0 > 1$, the uninfected steady state E_0 becomes unstable and there is a unique infected steady state $E^+ = (H^+, I^+, D^+, V^+)$ which is locally asymptotically stable in the interior of Ω, where

$$H^+ = \frac{c\delta(\alpha+\beta)}{a\beta k},\ V^+ = \frac{a\beta s}{c\delta(\alpha+\beta)} - \frac{\mu}{k},\ D^+ = \frac{c}{\beta}V^+ \text{ and } I^+ = \frac{\alpha+\beta}{a}D^+.$$

On the other hand, it is supposed in system (1.1) that the individuals live in a constant environment. However, some parameters involved in epidemic models are inevitably affected by the environmental noise. There has been a growing interest in taking into

Table 1 Summary of parameter meaning of system (1.1)

Parameters	Descriptions
s	Production rate of the uninfected hepatocytes
k	Infection rate of the virions
μ	Natural death rate of the uninfected hepatocytes
δ	Clearance rate of the infected hepatocytes
a	Production rate of intracellular HBV DNA-containing capsids
β	Transmission rate of the capsids
α	Natural death rate of the capsids
c	Natural death rate of the virions

account the environmental noise in epidemiology model since it has turned out that the stochastic model can describe biological phenomena and viral infectious diseases in a more exact way (Jiang et al. 2017; Rihan and Alsakaji 2021; Hussain et al. 2022; Liu et al. 2017; Djordjevic et al. 2018; Lu et al. 2021). So far, there are different ways to introduce stochastic perturbations in the model. One of the most important ways is to assume that environmental fluctuations are of white noise type which mainly disturb the natural death rates μ, δ, α and c, respectively. More precisely, we can replace the rates μ, δ, α and c by average values plus random fluctuation terms, that is,

$$\begin{aligned} -\mu &\to -\mu + \sigma_1 \dot{B}_1(t), \quad -\delta \to -\delta + \sigma_2 \dot{B}_2(t), \\ -\alpha &\to -\alpha + \sigma_3 \dot{B}_3(t), \quad -c \to -c + \sigma_4 \dot{B}_4(t), \end{aligned}$$

where $B_1(t)$, $B_2(t)$, $B_3(t)$ and $B_4(t)$ represent mutually independent standard Brownian motions defined on a complete probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ with a filtration $\{\mathcal{F}_t\}_{t\geq 0}$ satisfying the usual conditions (Mao 1997), $\sigma_i^2 > 0$ represent the intensity of the white noises $\dot{B}_i(t)$ $(i = 1, 2, 3, 4)$, respectively. Then corresponding to the deterministic model (1.1), the stochastic system takes the following form:

$$\begin{cases} dH(t) = [s - kH(t)V(t) - \mu H(t)]dt + \sigma_1 H(t)dB_1(t), \\ dI(t) = [kH(t)V(t) - \delta I(t)]dt + \sigma_2 I(t)dB_2(t), \\ dD(t) = [aI(t) - \beta D(t) - \alpha D(t)]dt + \sigma_3 D(t)dB_3(t), \\ dV(t) = [\beta D(t) - cV(t)]dt + \sigma_4 V(t)dB_4(t). \end{cases} \tag{1.2}$$

Throughout this paper, denote by $C^2(\mathbb{R}^d; \overline{\mathbb{R}}_+)$ the family of all nonnegative functions $V(x)$ defined on $\mathbb{R}^d$ such that they are continuously twice differentiable in x. Let $\mathbf{I}_{\mathbb{A}}$ be the indicator function of the set $\mathbb{A}$. If G is a vector or matrix, we use the notation $\|G\|$ to represent its norm and its transpose is defined by G^{T}. If G is an invertible matrix, we use the notation G^{-1} to denote its inverse matrix. If G is a square matrix, its determinant is denoted by $|G|$. Moreover, we introduce the following notations:

$$\mathbb{R}_+^d = \{x = (x_1, \ldots, x_d)^{\mathrm{T}} \in \mathbb{R}^d : x_i > 0, 1 \leq i \leq d\},$$

$$\overline{\mathbb{R}}_+^d = \{x = (x_1, \ldots, x_d)^{\mathrm{T}} \in \mathbb{R}^d : x_i \geq 0, 1 \leq i \leq d\},$$
$$x \wedge y = \min\{x, y\},\ x \vee y = \max\{x, y\} \text{ for any } x, y \in \mathbb{R}.$$

This paper is organized as follows: In Sect. 2, we verify that there is a unique global positive solution to system (1.2) with any positive initial value. In Sect. 3, we adopt a stochastic Lyapunov function method to establish sufficient conditions for the existence and uniqueness of an ergodic stationary distribution of the proposed stochastic model (1.2). In Sect. 4, under the same conditions as Theorem 3.1, we obtain the accurate expression of probability density around the quasi-infected steady state of the stochastic system (1.2), which reflects the strong persistence of the infected hepatocytes. In Sect. 5, we establish sufficient conditions for extinction of the infected hepatocytes. In Sect. 6, numerical simulations are performed to support our analytical findings obtained in Sects. 3, 4 and 5. Finally, a brief conclusion is given to end this paper.

2 Existence and Uniqueness of the Global Positive Solution

Before studying the dynamical behavior of a stochastic model for a within host infection, we should ensure that the solution is global and positive. The following theorem guarantees the existence and uniqueness of the global positive solution of system (1.2) with any positive initial value.

Theorem 2.1 *For any initial value* $(H(0), I(0), D(0), V(0))^{\mathrm{T}} \in \mathbb{R}_+^4$*, system* (1.2) *has a unique solution* $(H(t), I(t), D(t), V(t))^{\mathrm{T}}$ *on* $t \geq 0$ *and the solution will remain in* $\mathbb{R}_+^4$ *with probability one, namely,* $(H(t), I(t), D(t), V(t))^{\mathrm{T}} \in \mathbb{R}_+^4$ *for all* $t \geq 0$ *almost surely (a.s.).*

Proof Since the coefficients of system (1.2) are locally Lipschitz continuous, then for any initial value $(H(0), I(0), D(0), V(0))^{\mathrm{T}} \in \mathbb{R}_+^4$, there is a unique local solution $(H(t), I(t), D(t), V(t))^{\mathrm{T}} \in \mathbb{R}_+^4$ to system (1.2) on $[0, \tau_e)$ a.s., where τ_e is the explosion time (Mao 1997). To show the solution is global which is equivalent to proving $\tau_e = \infty$ a.s. We will show the solution will not explode to infinity in a finite time. To this end, let $n_0 \geq 1$ be sufficiently large such that $H(0)$, $I(0)$, $D(0)$ and $V(0)$ all lie within the interval $[1/n_0, n_0]$. For each integer $n \geq n_0$, we define a stopping time by Mao (1997)

$$\tau_n = \inf\Big\{t \in [0, \tau_e) : \min\{H(t), I(t), D(t), V(t)\} \leq \frac{1}{n} \text{ or } \max\{H(t), I(t), D(t), V(t)\} \geq n\Big\},$$

where throughout this paper we set $\inf \emptyset = \infty$ (commonly, $\emptyset$ represents the empty set). According to the definition, we can see that τ_n is increasing as $n \to \infty$. Denote by $\tau_\infty = \lim_{n\to\infty} \tau_n$, whence $\tau_\infty \leq \tau_e$ a.s. If $\tau_\infty = \infty$ a.s. holds, then $\tau_e = \infty$ a.s. and $(H(t), I(t), D(t), V(t))^{\mathrm{T}} \in \mathbb{R}_+^4$ a.s. for all $t \geq 0$. If this statement is false, then

there is a pair of constants $T > 0$ and $\epsilon \in (0, 1)$ such that

$$\mathbb{P}\{\tau_\infty \leq T\} > \epsilon.$$

Accordingly, there exists an integer $n_1 \geq n_0$ such that

$$\mathbb{P}\{\tau_n \leq T\} \geq \epsilon \text{ for all } n \geq n_1.$$

For any $(H, I, D, V) \in \mathbb{R}^4_+$, a nonnegative C^2-function U is defined by

$$\begin{aligned} U(H, I, D, V) = &\left(H - b_1 - b_1 \ln \frac{H}{b_1}\right) + (I - 1 - \ln I) \\ &+ b_2(D - 1 - \ln D) + b_3(V - 1 - \ln V), \end{aligned}$$

where b_1, b_2, b_3 are positive constants which will be determined later. It is noticed that the nonnegativity of above function is ensured since $u - 1 - \ln u \geq 0$ for any $u > 0$. Applying Itô's formula (Mao 1997) to U leads to that

$$\begin{aligned} dU(H, I, D, V) = &LU(H, I, D, V)dt + \sigma_1(H - b_1)dB_1(t) + \sigma_2(I - 1)dB_2(t) \\ &+ b_2\sigma_3(D - 1)dB_3(t) + b_3\sigma_4(V - 1)dB_4(t), \end{aligned}$$

where $LU : \mathbb{R}^4_+ \to \mathbb{R}$ is defined by

$$\begin{aligned} LU = &s - kHV - \mu H - \frac{b_1 s}{H} + b_1 kV + b_1\mu + \frac{b_1\sigma_1^2}{2} \\ &+ kHV - \delta I - \frac{kHV}{I} + \delta + \frac{\sigma_2^2}{2} + b_2 aI - b_2(\alpha + \beta)D - \frac{ab_2 I}{D} \\ &+ b_2(\alpha + \beta) + \frac{b_2\sigma_3^2}{2} + b_3\beta D - b_3 cV - \frac{b_3\beta D}{V} + b_3 c + \frac{b_3\sigma_4^2}{2} \\ \leq &s + b_1\mu + \delta + b_2(\alpha + \beta) + b_3 c + \frac{b_1\sigma_1^2}{2} + \frac{\sigma_2^2}{2} \\ &+ \frac{b_2\sigma_3^2}{2} + \frac{b_3\sigma_4^2}{2} + (b_2 a - \delta)I + [b_3\beta - b_2(\alpha + \beta)]D + (b_1 k - b_3 c)V. \end{aligned}$$

Choose $b_1 = \delta c(\alpha+\beta)/(a\beta k)$, $b_2 = \delta/a$, $b_3 = \delta(\alpha+\beta)/(a\beta)$ such that $b_2 a - \delta = 0$, $b_3\beta - b_2(\alpha + \beta) = 0$, $b_1 k - b_3 c = 0$, then we have

$$\begin{aligned} LU \leq &s + b_1\mu + \delta + b_2(\alpha + \beta) + b_3 c + \frac{b_1\sigma_1^2}{2} + \frac{\sigma_2^2}{2} + \frac{b_2\sigma_3^2}{2} + \frac{b_3\sigma_4^2}{2} \\ := &K. \end{aligned}$$

Here K is a positive constant which is independent of the variables H, I, D and V. The remainder of the proof is similar to Theorem 2.1 in the literature (Lu et al. 2021) and hence it is omitted here. This completes the proof. □

3 Existence of Ergodic Stationary Distribution

In this section, we mainly focus on the existence and uniqueness of an ergodic stationary distribution, which implies the persistence of uninfected hepatocytes, infected hepatocytes, DNA-containing capsids and the virions. We will use the stochastic Lyapunov function method to make it. Suppose X is a random process. If X is a Markov chain, then the stochastic Lyapunov function is a function for which an operator is defined as a limit function. We first introduce some theories about stochastic differential equations and stationary distribution.

Let $X(t)$ be a regular time-homogeneous Markov process in $\mathbb{R}^d$ described by the general stochastic differential equation

$$dX(t) = f(X(t))dt + g(X(t))dB(t), \quad t \geq t_0, \ X \in \mathbb{R}^d, \tag{3.1}$$

with the initial value $X(t_0) = X_0 \in \mathbb{R}^d$, where $B(t)$ denotes a d-dimensional standard Brownian motion defined on a complete filtered probability space $(\Omega, \mathcal{F}, \mathcal{F}_t^B, \mathbb{P})$, where $\mathcal{F}_t^B$ is the filtration generated by it up to time t and satisfies the usual conditions (Mao 1997). Denote by $C^2(\mathbb{R}^d; \overline{\mathbb{R}}_+)$ the family of all nonnegative functions $V(X)$ defined on $\mathbb{R}^d$ such that they are continuously twice differentiable in X.

The solution $X(t)$ is unique and global according to theorems considered in Mao (1997). The two functions f, g are assumed to satisfy the local Lipschitz condition, where f is defined on $\mathbb{R}^d$, and g is a $d \times m$ matrix.

Define the differential operator L associated with (3.1) by

$$L = \sum_{i=1}^{d} f_i \frac{\partial}{\partial X_i} + \frac{1}{2} \sum_{i,j=1}^{d} \left[g^{\mathrm{T}} \cdot g\right]_{ij} \frac{\partial^2}{\partial X_i \partial X_j}.$$

This operator can be used to a C^2-Lyapunov function $V : \mathbb{R}^d \to \overline{\mathbb{R}}_+$, then

$$LV(X) = V_X(X) f(X) + \frac{1}{2} \mathrm{tr}\left[g^{\mathrm{T}}(X) V_{XX}(X) g(X)\right],$$

where $V_X = (\partial V/\partial X_1, \ldots, \partial V/\partial X_d)$, $V_{XX} = (\partial^2 V/(\partial X_i \partial X_j))_{d\times d}$ and the notation tr denotes the trace of the matrix.

In addition, the diffusion matrix of the process $X(t)$ is defined as follows:

$$A(x) = (a_{ij}(x)), \ a_{ij}(x) = \sum_{r=1}^{k} g_r^i(x) g_r^j(x).$$

Lemma 3.1 (Mao 1997) *Itô's formula is the chain rule in Itô calculus which gives an expression to the differential of $V(X)$. Then in view of (3.1), we obtain*

$$dV(t) = LV(X(t))dt + V_X(X(t))g(X(t))dB(t).$$

Lemma 3.2 (Khasminskii 2012) *The Markov process $X(t)$ has a unique ergodic stationary distribution $\pi(\cdot)$ if there exists a bounded open domain $U \subset \mathbb{R}^d$ with regular boundary Γ, having the following properties:*

(A_1) *In the domain U and some neighborhood thereof, the smallest eigenvalue of the diffusion matrix $A(x)$ is bounded away from zero.*

(A_2) *If $x \in \mathbb{R}^d \backslash U$, the mean time τ at which a path issuing from x reaches the set U is finite, and $\sup_{x\in D} \mathbb{E}^x \tau < \infty$ for every compact subset $D \subset \mathbb{R}^d$.*

Remark 3.1 To validate A_1, it needs to verify that the operator F is uniformly elliptical in U, where $Fu = b(x)u_x + (\mathrm{tr}(A(x)u_{xx}))/2$, that is, there is a positive number $\mathbb{M}$ such that

$$\sum_{i,j=1}^{d} a_{ij}(x)\xi_i\xi_j \geq \mathbb{M}\|\xi\|^2, \quad x \in U, \ \xi \in \mathbb{R}^d$$

[see Chapter 3, p. 103 of Gard (1988) and Rayleigh's principle in Strang (1988), Chapter 6, p. 349]. To prove A_2, it suffices show that there exist some neighborhood U and a nonnegative C^2-function such that for any $\mathbb{R}^d \backslash U$, LV is negative [for details the readers can refer to Zhu and Yin (2007, p. 1163)].

Theorem 3.1 *Assume that*

$$R_0^S := \frac{a\beta sk}{\left(\mu + \frac{\sigma_1^2}{2}\right)\left(\delta + \frac{\sigma_2^2}{2}\right)\left(\alpha + \beta + \frac{\sigma_3^2}{2}\right)\left(c + \frac{\sigma_4^2}{2}\right)} > 1,$$

then there is a positive solution $(H(t), I(t), D(t), V(t))^{\mathrm{T}}$ of system (1.2) which has a unique stationary distribution $\pi(\cdot)$ and the ergodicity holds.

Proof The proof process is divided into two steps: the first step is to prove that the uniform elliptic condition is satisfied, and the second step is to construct a nonnegative Lyapunov function which satisfies the condition (A_2) of Lemma 3.2.

Step 1 The diffusion matrix of system (1.2) is given by

$$A = \begin{pmatrix} \sigma_1^2 H^2 & 0 & 0 & 0 \\ 0 & \sigma_2^2 I^2 & 0 & 0 \\ 0 & 0 & \sigma_3^2 D^2 & 0 \\ 0 & 0 & 0 & \sigma_4^2 V^2 \end{pmatrix}.$$

Choose $\mathbb{M} = \min_{(H,I,D,V)^{\mathrm{T}} \in \bar{U}_n \subset \mathbb{R}_+^4} \{\sigma_1^2 H^2, \sigma_2^2 I^2, \sigma_3^2 D^2, \sigma_4^2 V^2\}$, we have

$$\sum_{i,j=1}^{4} a_{ij}(H, I, D, V)\zeta_i\zeta_j = \sigma_1^2 H^2\zeta_1^2 + \sigma_2^2 I^2\zeta_2^2 + \sigma_3^2 D^2\zeta_3^2 + \sigma_4^2 V^2\zeta_4^2 \geq \mathbb{M}\|\zeta\|^2,$$

for any $(H, I, D, V)^{\mathrm{T}} \in \bar{U}_n$ and $\zeta = (\zeta_1, \zeta_2, \zeta_3, \zeta_4)^{\mathrm{T}} \in \mathbb{R}^4$, where $\bar{U}_n = [1/n, n] \times [1/n, n] \times [1/n, n] \times [1/n, n]$. Accordingly, the condition (A_1) of Lemma 3.2 holds.

Step 2 Let

$$\bar{H} = \frac{s}{\mu + \frac{\sigma_1^2}{2}},\ \bar{I} = 1,\ \bar{D} = \frac{a\bar{I}}{\alpha + \beta + \frac{\sigma_3^2}{2}},\ \bar{V} = \frac{\beta\bar{D}}{c + \frac{\sigma_4^2}{2}}, \tilde{H} = \frac{H}{\bar{H}},\ \tilde{I} = \frac{I}{\bar{I}},$$
$$\tilde{D} = \frac{D}{\bar{D}},\ \tilde{V} = \frac{V}{\bar{V}}.$$

In view of system (1.2), we obtain

$$\begin{aligned} L(-\ln H) &= -\frac{s}{\bar{H}}\frac{1}{\tilde{H}} + \mu + \frac{\sigma_1^2}{2} + kV \\ &\le -\frac{s}{\bar{H}}\left(\ln\frac{1}{\tilde{H}} + 1\right) + \mu + \frac{\sigma_1^2}{2} + kV \\ &= \frac{s}{\bar{H}}\ln\tilde{H} + kV, \end{aligned} \tag{3.2}$$

$$\begin{aligned} L(-\ln I) &= -\frac{k\bar{H}\bar{V}}{\bar{I}}\frac{\tilde{H}\tilde{V}}{\tilde{I}} + \delta + \frac{\sigma_2^2}{2} \\ &\le -\frac{k\bar{H}\bar{V}}{\bar{I}}\left(\ln\frac{\tilde{H}\tilde{V}}{\tilde{I}} + 1\right) + \delta + \frac{\sigma_2^2}{2} \\ &= -k\bar{H}\bar{V} + \delta + \frac{\sigma_2^2}{2} + k\bar{H}\bar{V}\ln\tilde{I} - k\bar{H}\bar{V}\ln\tilde{H} - k\bar{H}\bar{V}\ln\tilde{V}, \end{aligned} \tag{3.3}$$

$$\begin{aligned} L(-\ln D) &= -\frac{a\bar{I}}{\bar{D}}\frac{\tilde{I}}{\tilde{D}} + \alpha + \beta + \frac{\sigma_3^2}{2} \\ &\le -\frac{a\bar{I}}{\bar{D}}\left(\ln\frac{\tilde{I}}{\tilde{D}} + 1\right) + \alpha + \beta + \frac{\sigma_3^2}{2} \\ &= -\frac{a\bar{I}}{\bar{D}}\ln\tilde{I} + \frac{a\bar{I}}{\bar{D}}\ln\tilde{D}, \end{aligned} \tag{3.4}$$

and

$$\begin{aligned} L(-\ln V) &= -\frac{\beta\bar{D}}{\bar{V}}\frac{\tilde{D}}{\tilde{V}} + c + \frac{\sigma_4^2}{2} \\ &\le -\frac{\beta\bar{D}}{\bar{V}}\left(\ln\frac{\tilde{D}}{\tilde{V}} + 1\right) + c + \frac{\sigma_4^2}{2} \\ &= -\frac{\beta\bar{D}}{\bar{V}}\ln\tilde{D} + \frac{\beta\bar{D}}{\bar{V}}\ln\tilde{V}. \end{aligned} \tag{3.5}$$

Define

$$U_1(H, I, D, V) = -\ln I - c_1 \ln H - c_2 \ln D - c_3 \ln V + \frac{c_1 k}{c} V + \frac{c_1 k \beta}{c(\alpha+\beta)} D,$$

where c_1, c_2 and c_3 are positive constants which will be determined later. Then, from (3.2), (3.3), (3.4) and (3.5), it follows that

$$\begin{aligned}
LU_1 \leq & -k\bar{H}\bar{V} + \delta + \frac{\sigma_2^2}{2} + k\bar{H}\bar{V}\ln\tilde{I} - k\bar{H}\bar{V}\ln\tilde{H} - k\bar{H}\bar{V}\ln\tilde{V} \\
& + \frac{c_1 s}{\bar{H}}\ln\tilde{H} - \frac{c_2 a\bar{I}}{\bar{D}}\ln\tilde{I} + \frac{c_2 a\bar{I}}{\bar{D}}\ln\tilde{D} - \frac{c_3\beta\bar{D}}{\bar{V}}\ln\tilde{D} \\
& + \frac{c_3\beta\bar{D}}{\bar{V}}\ln\tilde{V} + c_1 kV + \frac{c_1 k}{c}(\beta D - cV) + \frac{c_1 k\beta}{c(\alpha+\beta)}(aI - \beta D - \alpha D) \\
= & -k\bar{H}\bar{V} + \delta + \frac{\sigma_2^2}{2} + \left(\frac{c_1 s}{\bar{H}} - k\bar{H}\bar{V}\right)\ln\tilde{H} \\
& + \left(k\bar{H}\bar{V} - \frac{c_2 a\bar{I}}{\bar{D}}\right)\ln\tilde{I} + \left(\frac{c_2 a\bar{I}}{\bar{D}} - \frac{c_3\beta\bar{D}}{\bar{V}}\right)\ln\tilde{D} + \left(\frac{c_3\beta\bar{D}}{\bar{V}} - k\bar{H}\bar{V}\right)\ln\tilde{V} \\
& + \frac{c_1 k\beta a}{c(\alpha+\beta)} I.
\end{aligned}$$

Let

$$\begin{cases}
\dfrac{c_1 s}{\bar{H}} - k\bar{H}\bar{V} = 0, \\
\dfrac{c_2 a\bar{I}}{\bar{D}} - k\bar{H}\bar{V} = 0, \\
\dfrac{c_3\beta\bar{D}}{\bar{V}} - k\bar{H}\bar{V} = 0,
\end{cases}$$

we obtain

$$\begin{cases}
c_1 = \dfrac{k\bar{H}^2\bar{V}}{s}, \\
c_2 = \dfrac{k\bar{H}\bar{D}\bar{V}}{a}, \\
c_3 = \dfrac{k\bar{H}\bar{V}^2}{\beta\bar{D}}.
\end{cases}$$

Therefore

$$\begin{aligned}
LU_1 &\leq -k\bar{H}\bar{V} + \delta + \frac{\sigma_2^2}{2} + \frac{c_1 k\beta a}{c(\alpha+\beta)} I \\
&= -\left(\delta + \frac{\sigma_2^2}{2}\right)\left(R_0^S - 1\right) + \frac{c_1 k\beta a}{c(\alpha+\beta)} I,
\end{aligned} \tag{3.6}$$

where

$$R_0^S := \frac{k\bar{H}\bar{V}}{\delta + \frac{\sigma_2^2}{2}} = \frac{a\beta sk}{\left(\mu + \frac{\sigma_1^2}{2}\right)\left(\delta + \frac{\sigma_2^2}{2}\right)\left(\alpha + \beta + \frac{\sigma_3^2}{2}\right)\left(c + \frac{\sigma_4^2}{2}\right)}.$$

Next, define

$$\begin{aligned} U_2(H) &= -\ln H,\ U_3(D) = -\ln D,\ U_4(V) = -\ln V,\ U_5(H, I, D, V) \\ &= \frac{1}{\theta+1}\left(H + I + \frac{\delta}{2a}D + \frac{\delta(\alpha+\beta)}{4a\beta}V\right)^{\theta+1}, \end{aligned}$$

where $\theta \in (0, (2\mu \wedge \delta \wedge (\alpha+\beta) \wedge 2c)/(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2))$ is a sufficiently small constant. Applying Itô's formula to U_2, U_3, U_4 and U_5, respectively, we have

$$LU_2 = -\frac{s}{H} + kV + \mu + \frac{\sigma_1^2}{2}, \tag{3.7}$$

$$LU_3 = -\frac{aI}{D} + \alpha + \beta + \frac{\sigma_3^2}{2}, \tag{3.8}$$

$$LU_4 = -\frac{\beta D}{V} + c + \frac{\sigma_4^2}{2}, \tag{3.9}$$

and

$$\begin{aligned} LU_5 = &\left(H + I + \frac{\delta}{2a}D + \frac{\delta(\alpha+\beta)}{4a\beta}V\right)^{\theta} \\ &\left(s - \mu H - \frac{\delta}{2}I - \frac{\delta(\alpha+\beta)}{4a}D - \frac{\delta c(\alpha+\beta)}{4a\beta}V\right) \\ &+ \frac{\theta}{2}\left(H + I + \frac{\delta}{2a}D + \frac{\delta(\alpha+\beta)}{4a\beta}V\right)^{\theta-1} \\ &\left(\sigma_1^2 H^2 + \sigma_2^2 I^2 + \frac{\delta^2\sigma_3^2}{(2a)^2}D^2 + \frac{\delta^2(\alpha+\beta)^2\sigma_4^2}{(4a\beta)^2}V^2\right) \\ \le\ & s\left(H + I + \frac{\delta}{2a}D + \frac{\delta(\alpha+\beta)}{4a\beta}V\right)^{\theta} \\ &- \left(\mu \wedge \frac{\delta}{2} \wedge \frac{\alpha+\beta}{2} \wedge c\right)\left(H + I + \frac{\delta}{2a}D + \frac{\delta(\alpha+\beta)}{4a\beta}V\right)^{\theta+1} \\ &+ \frac{\theta}{2}(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2)\left(H + I + \frac{\delta}{2a}D + \frac{\delta(\alpha+\beta)}{4a\beta}V\right)^{\theta+1} \\ \le\ & \mathbb{J}_1 - \frac{1}{2}\left[\left(\mu \wedge \frac{\delta}{2} \wedge \frac{\alpha+\beta}{2} \wedge c\right) - \frac{\theta}{2}(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2)\right] \\ &\left(H + I + \frac{\delta}{2a}D + \frac{\delta(\alpha+\beta)}{4a\beta}V\right)^{\theta+1} \end{aligned}$$

$$
\begin{aligned}
&\leq \mathbb{J}_1 - \frac{1}{2}\left[\left(\mu \wedge \frac{\delta}{2} \wedge \frac{\alpha+\beta}{2} \wedge c\right) - \frac{\theta}{2}(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2)\right] \\
&\quad \left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) \\
&= \mathbb{J}_1 - \frac{\vartheta}{2}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right), \quad (3.10)
\end{aligned}
$$

where

$$
\begin{aligned}
\mathbb{J}_1 := \sup_{(H,I,D,V)^{\mathrm{T}} \in \mathbb{R}_+^4} &\left\{ s\left(H + I + \frac{\delta}{2a} D + \frac{\delta(\alpha+\beta)}{4a\beta} V\right)^{\theta} \right. \\
&\left. - \frac{\vartheta}{2}\left(H + I + \frac{\delta}{2a} D + \frac{\delta(\alpha+\beta)}{4a\beta} V\right)^{\theta+1} \right\} < \infty,
\end{aligned}
$$

and

$$
\vartheta := \left(\mu \wedge \frac{\delta}{2} \wedge \frac{\alpha+\beta}{2} \wedge c\right) - \frac{\theta}{2}\left(\sigma_1^2 \vee \sigma_2^2 \vee \sigma_3^2 \vee \sigma_4^2\right).
$$

Define a C^2-function $U : \mathbb{R}_+^4 \to \mathbb{R}$ in the following form

$$
\begin{aligned}
&U(H, I, D, V) \\
&\quad = MU_1(H, I, D, V) + U_2(H) + U_3(D) + U_4(V) + U_5(H, I, D, V),
\end{aligned}
$$

where M is a sufficiently large positive constant satisfying

$$
-M\left(\delta + \frac{\sigma_2^2}{2}\right)\left(R_0^S - 1\right) + \mathbb{J}_2 \leq -2, \quad (3.11)
$$

and

$$
\begin{aligned}
\mathbb{J}_2 := \sup_{(H,I,D,V)^{\mathrm{T}} \in \mathbb{R}_+^4} &\left\{ -\frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) \right. \\
&\left. + kV + \mathbb{J}_1 + \mu + \alpha + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \right\} < \infty.
\end{aligned}
$$

It is noticed that $U(H, I, D, V)$ is not only continuous, but also tends to ∞ as $(H, I, D, V)^{\mathrm{T}}$ approaches the boundary of $\mathbb{R}_+^4$. Therefore, it must have a lower bound and reach this lower bound at a point $(H^0, I^0, D^0, V^0)^{\mathrm{T}}$ in the interior of $\mathbb{R}_+^4$. Then,

we define a C^2-function $\bar{U} : \mathbb{R}_+^4 \to \overline{\mathbb{R}}_+$ as follows:

$$\begin{aligned}\bar{U}(H, I, D, V) &= U(H, I, D, V) - U(H^0, I^0, D^0, V^0)\\ &= MU_1(H, I, D, V) + U_2(H) + U_3(D) + U_4(V)\\ &\quad + U_5(H, I, D, V) - U(H^0, I^0, D^0, V^0).\end{aligned}$$

According to (3.6), (3.7), (3.8), (3.9) and (3), we have

$$\begin{aligned}L\bar{U} \le &-M\left(\delta + \frac{\sigma_2^2}{2}\right)\left(R_0^S - 1\right) + \frac{Mc_1k\beta a}{c(\alpha+\beta)}I - \frac{s}{H} - \frac{\vartheta}{2}\\ &\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}}D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}}V^{\theta+1}\right)\\ &- \frac{aI}{D} - \frac{\beta D}{V} + kV + \mathbb{J}_1 + \mu + \alpha + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2}.\end{aligned} \tag{3.12}$$

Next, we construct a compact subset U_ϵ such that the condition (A_2) of Lemma 3.1 holds. The bounded closed set is chosen as follows:

$$U_\epsilon = \left\{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : \epsilon \le H \le \frac{1}{\epsilon}, \epsilon \le I \le \frac{1}{\epsilon}, \epsilon^2 \le D \le \frac{1}{\epsilon^2}, \epsilon^3 \le V \le \frac{1}{\epsilon^3}\right\},$$

where $0 < \epsilon < 1$ is a sufficiently small constant. In the set $\mathbb{R}_+^4 \backslash U_\epsilon$, we can select ϵ small enough to satisfy the following conditions,

$$-\frac{s}{\epsilon} + \mathbb{J}_3 \le -1, \tag{3.13}$$

$$\epsilon \le \frac{c(\alpha+\beta)}{Mc_1k\beta a}, \tag{3.14}$$

$$-\frac{a}{\epsilon} + \mathbb{J}_3 \le -1, \tag{3.15}$$

$$-\frac{\beta}{\epsilon} + \mathbb{J}_3 \le -1, \tag{3.16}$$

$$-\frac{\vartheta}{4\epsilon^{\theta+1}} + \mathbb{J}_3 \le -1, \tag{3.17}$$

$$-\frac{\vartheta\delta^{\theta+1}}{2^{\theta+3}a^{\theta+1}\epsilon^{2(\theta+1)}} + \mathbb{J}_3 \le -1, \tag{3.18}$$

$$-\frac{\vartheta\delta^{\theta+1}(\alpha+\beta)^{\theta+1}}{4^{\theta+2}(a\beta)^{\theta+1}\epsilon^{3(\theta+1)}} + \mathbb{J}_3 \le -1. \tag{3.19}$$

Here, J_3 is a positive constant giving explicitly in the expression (3.21). For convenience, we can split $\mathbb{R}_+^4 \backslash U_\epsilon$ into the following eight domains,

$$U_1 = \{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : H < \epsilon\},\ U_2 = \{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : I < \epsilon\},$$

$$U_3 = \{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : I \geq \epsilon, D < \epsilon^2\},$$
$$U_4 = \{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : D \geq \epsilon^2, V < \epsilon^3\},$$
$$U_5 = \left\{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : H > \frac{1}{\epsilon}\right\}, U_6 = \left\{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : I > \frac{1}{\epsilon}\right\},$$
$$U_7 = \left\{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : D > \frac{1}{\epsilon^2}\right\}, \ U_8 = \left\{(H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 : V > \frac{1}{\epsilon^3}\right\}.$$

Apparently, $\mathbb{R}_+^4 \backslash U_\epsilon = U_1 \bigcup \ldots \bigcup U_8$. Next, we will show that $L\bar{U}(H, I, D, V) \leq -1$ on U_ϵ^c, that is, we need to prove its satisfaction on the eight domains.

Case 1. If $(H, I, D, V)^{\mathrm{T}} \in U_1$, according to (3.12), we have

$$\begin{aligned} L\bar{U} \leq & -\frac{s}{H} + \frac{Mc_1k\beta a}{c(\alpha+\beta)} I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) \\ & + kV + \mathbb{J}_1 + \mu + \alpha + \beta \\ & + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -\frac{s}{\epsilon} + \mathbb{J}_3 \\ \leq & -1, \end{aligned} \tag{3.20}$$

which follows from (3.13) and

$$\begin{aligned} \mathbb{J}_3 := \sup_{(H,I,D,V)^{\mathrm{T}} \in \mathbb{R}_+^4} & \left\{ -\frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) \right. \\ & + \frac{Mc_1k\beta a}{c(\alpha+\beta)} I + kV + \mathbb{J}_1 + \mu \\ & \left. + \alpha + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \right\} < \infty. \end{aligned} \tag{3.21}$$

Case 2. If $(H, I, D, V)^{\mathrm{T}} \in U_2$, by means of (3.12), we get

$$\begin{aligned} L\bar{U} \leq & -M\left(\delta + \frac{\sigma_2^2}{2}\right)(R_0^S - 1) + \frac{Mc_1k\beta a}{c(\alpha+\beta)} I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) + kV \\ & + \mathbb{J}_1 + \mu + \alpha + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -M\left(\delta + \frac{\sigma_2^2}{2}\right)(R_0^S - 1) + \frac{Mc_1k\beta a}{c(\alpha+\beta)} \epsilon + \mathbb{J}_2 \\ \leq & -1, \end{aligned} \tag{3.22}$$

which follows from (3.11) and (3.14) and $\mathbb{J}_2$ is defined in (3.11).

Case 3. If $(H, I, D, V)^{\mathrm{T}} \in U_3$, in view of (3.12), we obtain

$$\begin{aligned} L\bar{U} \leq & -\frac{aI}{D} + \frac{Mc_1k\beta a}{c(\alpha+\beta)}I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}}D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}}V^{\theta+1}\right) \\ & + kV + \mathbb{J}_1 + \mu + \alpha + \beta \\ & + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -\frac{a\epsilon}{\epsilon^2} + \mathbb{J}_3 \\ \leq & -1, \end{aligned} \tag{3.23}$$

which follows from (3.15).

Case 4. If $(H, I, D, V)^{\mathrm{T}} \in U_4$, by (3.12), we derive

$$\begin{aligned} L\bar{U} \leq & -\frac{\beta D}{V} + \frac{Mc_1k\beta a}{c(\alpha+\beta)}I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}}D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}}V^{\theta+1}\right) \\ & + kV + \mathbb{J}_1 + \mu + \alpha + \beta \\ & + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -\frac{\beta\epsilon^2}{\epsilon^3} + \mathbb{J}_3 \\ \leq & -1, \end{aligned} \tag{3.24}$$

which follows from (3.16).

Case 5. For any $(H, I, D, V)^{\mathrm{T}} \in U_5$, according to (3.12), we have

$$\begin{aligned} L\bar{U} \leq & -\frac{\vartheta}{4}H^{\theta+1} + \frac{Mc_1k\beta a}{c(\alpha+\beta)}I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}}D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}}V^{\theta+1}\right) \\ & + kV + \mathbb{J}_1 + \mu + \alpha \\ & + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -\frac{\vartheta}{4\epsilon^{\theta+1}} + \mathbb{J}_3 \\ \leq & -1, \end{aligned} \tag{3.25}$$

which follows from (3.17).

Case 6. If $(H, I, D, V)^{\mathrm{T}} \in U_6$, in view of (3.12), we obtain

$$\begin{aligned} L\bar{U} \leq & -\frac{\vartheta}{4} I^{\theta+1} + \frac{Mc_1 k\beta a}{c(\alpha+\beta)} I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) \\ & + kV + \mathbb{J}_1 + \mu + \alpha \\ & + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -\frac{\vartheta}{4\epsilon^{\theta+1}} + \mathbb{J}_3 \\ \leq & -1, \end{aligned} \tag{3.26}$$

which follows from (3.17).

Case 7. If $(H, I, D, V)^{\mathrm{T}} \in U_7$, by means of (3.12), we get

$$\begin{aligned} L\bar{U} \leq & -\frac{\vartheta\delta^{\theta+1}}{4(2a)^{\theta+1}} D^{\theta+1} + \frac{Mc_1 k\beta a}{c(\alpha+\beta)} I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) \\ & + kV + \mathbb{J}_1 + \mu \\ & + \alpha + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -\frac{\vartheta\delta^{\theta+1}}{2^{\theta+3} a^{\theta+1} \epsilon^{2(\theta+1)}} + \mathbb{J}_3 \\ \leq & -1, \end{aligned} \tag{3.27}$$

which follows from (3.18).

Case 8. If $(H, I, D, V)^{\mathrm{T}} \in U_8$, by (3.12), we derive

$$\begin{aligned} L\bar{U} \leq & -\frac{\vartheta\delta^{\theta+1}(\alpha+\beta)^{\theta+1}}{4(4a\beta)^{\theta+1}} V^{\theta+1} + \frac{Mc_1 k\beta a}{c(\alpha+\beta)} I \\ & - \frac{\vartheta}{4}\left(H^{\theta+1} + I^{\theta+1} + \frac{\delta^{\theta+1}}{(2a)^{\theta+1}} D^{\theta+1} + \frac{(\delta(\alpha+\beta))^{\theta+1}}{(4a\beta)^{\theta+1}} V^{\theta+1}\right) + kV \\ & + \mathbb{J}_1 + \mu + \alpha + \beta + c + \frac{\sigma_1^2}{2} + \frac{\sigma_3^2}{2} + \frac{\sigma_4^2}{2} \\ \leq & -\frac{\vartheta\delta^{\theta+1}(\alpha+\beta)^{\theta+1}}{4^{\theta+2}(a\beta)^{\theta+1}\epsilon^{3(\theta+1)}} + \mathbb{J}_3 \\ \leq & -1, \end{aligned} \tag{3.28}$$

which follows from (3.19).

Thus, according to (3.20), (3.22), (3.23), (3.24), (3.25), (3.26), (3.27) and (3.28), we can easily see that there exists a sufficiently small ϵ such that

$$L\bar{U} \leq -1 \text{ for any } (H, I, D, V)^{\mathrm{T}} \in \mathbb{R}_+^4 \backslash U_\epsilon,$$

which implies that the condition (A_2) of Lemma 3.2 also holds. In view of Lemma 3.2, we obtain that system (1.2) has a unique stationary distribution $\pi(\cdot)$ and the ergodicity holds. This completes the proof. □

4 Probability Density for System (1.2)

In view of Theorem 3.1, we get that the global positive solution $(H(t), I(t), D(t), V(t))^{\mathrm{T}}$ has a unique ergodic stationary distribution $\pi(\cdot)$ if $R_0^S > 1$. In this section, we will obtain the explicit expression of the probability density of the distribution $\pi(\cdot)$. Although there are lots of works that have used the Lyapunov equation [see (4.6) below] to find the probability distribution, such as the papers (Hayot and Jayaprakash 2004; Black et al. 2009; Pahle et al. 2012; Fatehi et al. 2018), the monograph of van Kampen (2007), and other works. But most of them have used numerical methods, while here we will present an analytical solution which is very interesting as there are few works on it. Firstly, two necessary transformations of system (1.2) are introduced.

4.1 Two Important Transformations of System (1.2)

(I) (Logarithmic transformation) Let $(u_1, u_2, u_3, u_4)^{\mathrm{T}} = (\ln H, \ln I, \ln D, \ln V)^{\mathrm{T}}$. Then by Itô's formula and system (1.2), we have

$$\begin{cases} du_1 = \left[se^{-u_1} - ke^{u_4} - \left(\mu + \dfrac{\sigma_1^2}{2}\right)\right]dt + \sigma_1 dB_1(t), \\ du_2 = \left[ke^{u_1+u_4-u_2} - \left(\delta + \dfrac{\sigma_2^2}{2}\right)\right]dt + \sigma_2 dB_2(t), \\ du_3 = \left[ae^{u_2-u_3} - \left(\beta + \alpha + \dfrac{\sigma_3^2}{2}\right)\right]dt + \sigma_3 dB_3(t), \\ du_4 = \left[\beta e^{u_3-u_4} - \left(c + \dfrac{\sigma_4^2}{2}\right)\right]dt + \sigma_4 dB_4(t). \end{cases} \tag{4.1}$$

Assume that $R_0^S > 1$, there is the quasi-infected steady state $E^*(H^*, I^*, D^*, V^*) = (e^{u_1^*}, e^{u_2^*}, e^{u_3^*}, e^{u_4^*})^{\mathrm{T}} \in \mathbb{R}_+^4$ determined by the following equations:

$$\begin{cases} se^{-u_1^*} - ke^{u_4^*} - \tilde{\mu} = 0, \\ ke^{u_1^*+u_4^*-u_2^*} - \tilde{\delta} = 0, \\ ae^{u_2^*-u_3^*} - (\beta + \tilde{\alpha}) = 0, \\ \beta e^{u_3^*-u_4^*} - \tilde{c} = 0, \end{cases} \tag{4.2}$$

where $\tilde{\mu} = \mu + \sigma_1^2/2$, $\tilde{\delta} = \delta + \sigma_2^2/2$, $\tilde{\alpha} = \alpha + \sigma_3^2/2$, $\tilde{c} = c + \sigma_4^2/2$. Thus, by (4.2), we can obtain that $H^* = (\tilde{c}\tilde{\delta}(\beta + \tilde{\alpha}))/(a\beta k)$, $V^* = (a\beta s)/(\tilde{c}\tilde{\delta}(\beta + \tilde{\alpha})) - \tilde{\mu}/k$, $D^* = (\tilde{c}V^*)/\beta$ and $I^* = (\beta + \tilde{\alpha})D^*/a$.

(II) (Equilibrium offset transformation) Let $X = (x_1, x_2, x_3, x_4)^{\mathrm{T}} = (u_1 - u_1^*, u_2 - u_2^*, u_3 - u_3^*, u_4 - u_4^*)^{\mathrm{T}}$, the corresponding linearized system of (4.1) is as follows:

$$\begin{cases} dx_1 = (-a_{11}x_1 - a_{14}x_4)dt + \sigma_1 dB_1(t), \\ dx_2 = (a_{21}x_1 - a_{21}x_2 + a_{21}x_4)dt + \sigma_2 dB_2(t), \\ dx_3 = (a_{32}x_2 - a_{32}x_3)dt + \sigma_3 dB_3(t), \\ dx_4 = (a_{43}x_3 - a_{43}x_4)dt + \sigma_4 dB_4(t), \end{cases} \tag{4.3}$$

where

$$a_{11} = se^{-u_1^*},\ a_{14} = ke^{u_4^*},\ a_{21} = ke^{u_1^* + u_4^* - u_2^*},\ a_{32} = ae^{u_2^* - u_3^*},\ a_{43} = \beta e^{u_3^* - u_4^*},$$

and $a_{11} > 0$, $a_{14} > 0$, $a_{21} > 0$, $a_{32} > 0$ and $a_{43} > 0$.

Before giving the accurate expression of probability density, we need to introduce an important definition and two necessary lemmas.

Definition 4.1 (Ma et al. 2015) The characteristic polynomial of the square matrix A_n is defined as $\varphi_{A_n}(\lambda) = \lambda^n + a_1\lambda^{n-1} + \cdots + a_{n-1}\lambda + a_n$, then A_n is called a Hurwitz matrix if and only if A_n has all negative real-part eigenvalues, i.e.,

$$H_k = \begin{vmatrix} a_1 & a_3 & a_5 & \dots & a_{2k-1} \\ 1 & a_2 & a_4 & \dots & a_{2k-2} \\ 0 & a_1 & a_3 & \dots & a_{2k-3} \\ 0 & 1 & a_2 & \dots & a_{2k-4} \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & a_k \end{vmatrix} > 0, \quad k = 1, 2, \dots, n,$$

where the complementary definition is $a_j = 0$, $j > n$. Additionally, the corresponding necessary conditions for A_n to be a Hurwitz matrix

$$(i)\ a_j > 0,\ \ j = 1, 2, \dots, n;\ \ (ii)\ a_i a_{i+1} > a_{i-1}a_{i+2},\ \ i = 1, 2, \dots, n-2\ (a_0 = 1).$$

Lemma 4.1 (Ge et al. 2022) *For the algebraic equation $H_0^2 + C_0\Sigma_0 + \Sigma_0 C_0^{\mathrm{T}} = 0$, where $H_0 = diag(1, 0, 0, 0)$ and Σ_0 is a real symmetric matrix, and the standard matrix*

$$C_0 = \begin{pmatrix} -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

If $\tau_1 > 0$, $\tau_3 > 0$, $\tau_4 > 0$ and $\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4 > 0$, then Σ_0 is a positive definite matrix, where

$$\Sigma_0 = \begin{pmatrix} \frac{\tau_2\tau_3 - \tau_1\tau_4}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & -\frac{\tau_3}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 \\ 0 & \frac{\tau_3}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} \\ -\frac{\tau_3}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & \frac{\tau_1}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 \\ 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & \frac{\tau_1\tau_2 - \tau_3}{2\tau_4(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} \end{pmatrix}.$$

Here C_0 in this form is called the standard R_1 matrix.

Lemma 4.2 (Ge et al. 2022) *For the algebraic equation $H_0^2 + \tilde{C}_0\Theta_0 + \Theta_0\tilde{C}_0^{\mathrm{T}} = 0$, where $H_0 = diag(1, 0, 0, 0)$, Θ_0 is a real symmetric matrix, and the standard matrix*

$$\tilde{C}_0 = \begin{pmatrix} -l_1 & -l_2 & -l_3 & -l_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -a_{21} \end{pmatrix}.$$

If $l_1 > 0$, $l_3 > 0$ and $l_1l_2 - l_3 > 0$, then the matrix Θ_0 is semi-positive definite which follows

$$\Theta_0 = \begin{pmatrix} \frac{l_2}{2(l_1l_2 - l_3)} & 0 & -\frac{1}{2(l_1l_2 - l_3)} & 0 \\ 0 & \frac{1}{2(l_1l_2 - l_3)} & 0 & 0 \\ -\frac{1}{2(l_1l_2 - l_3)} & 0 & \frac{l_1}{2l_3(l_1l_2 - l_3)} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Here $\tilde{C}_0$ in this form is called the standard R_2 matrix.

4.2 Probability Density of Stationary Distribution $\pi(\cdot)$

Theorem 4.1 *If $R_0^S > 1$, then for any initial value $(H(0), I(0), D(0), V(0))^{\mathrm{T}} \in \mathbb{R}_+^4$, the stationary distribution $\pi(\cdot)$ around E^* has a unique log-normal probability density $\Phi(H, I, D, V)$, which can be denoted by*

$$\Phi(H, I, D, V) = (2\pi)^{-2}|\Sigma|^{-\frac{1}{2}} e^{-\frac{1}{2}(\ln\frac{H}{H^*}, \ln\frac{I}{I^*}, \ln\frac{D}{D^*}, \ln\frac{V}{V^*})\Sigma^{-1}(\ln\frac{H}{H^*}, \ln\frac{I}{I^*}, \ln\frac{D}{D^*}, \ln\frac{V}{V^*})^{\mathrm{T}}},$$

where the covariance matrix Σ is positive definite, and the specific form of Σ is given as follows:

(1) *If* $m_1 = 0$, $m_2 = 0$, *then*

$$\begin{aligned}\Sigma &= \varrho_1^2 J_1^{-1}\Sigma_0\left(J_1^{-1}\right)^{\mathrm{T}} + \varrho_2^2\left(J_3J_2\right)^{-1}\Sigma_0\left[\left(J_3J_2\right)^{-1}\right]^{\mathrm{T}} \\ &\quad + \tilde{\varrho}_3^2\left(J_7J_5J_4\right)^{-1}\Theta_0\left[\left(J_7J_5J_4\right)^{-1}\right]^{\mathrm{T}} + \tilde{\varrho}_4^2\left(Q_7Q_6Q_2Q_1\right)^{-1}\Theta_0 \\ &\quad \left[\left(Q_7Q_6Q_2Q_1\right)^{-1}\right]^{\mathrm{T}}.\end{aligned}$$

(2) *If* $m_1 \neq 0$, $m_2 = 0$, *then*

$$\begin{aligned}\Sigma &= \varrho_1^2 J_1^{-1}\Sigma_0\left(J_1^{-1}\right)^{\mathrm{T}} + \varrho_2^2\left(J_3J_2\right)^{-1}\Sigma_0\left[\left(J_3J_2\right)^{-1}\right]^{\mathrm{T}} \\ &\quad + \varrho_3^2\left(J_6J_5J_4\right)^{-1}\Sigma_0\left[\left(J_6J_5J_4\right)^{-1}\right]^{\mathrm{T}} + \tilde{\varrho}_4^2\left(Q_7Q_6Q_2Q_1\right)^{-1}\Theta_0 \\ &\quad \left[\left(Q_7Q_6Q_2Q_1\right)^{-1}\right]^{\mathrm{T}}.\end{aligned}$$

(3) *If* $m_1 = 0$, $m_3 \neq 0$, *then*

$$\begin{aligned}\Sigma &= \varrho_1^2 J_1^{-1}\Sigma_0\left(J_1^{-1}\right)^{\mathrm{T}} + \varrho_2^2\left(J_3J_2\right)^{-1}\Sigma_0\left[\left(J_3J_2\right)^{-1}\right]^{\mathrm{T}} \\ &\quad + \tilde{\varrho}_3^2\left(J_7J_5J_4\right)^{-1}\Theta_0\left[\left(J_7J_5J_4\right)^{-1}\right]^{\mathrm{T}} + \varrho_4^2\left(Q_4Q_3Q_2Q_1\right)^{-1}\Sigma_0 \\ &\quad \left[\left(Q_4Q_3Q_2Q_1\right)^{-1}\right]^{\mathrm{T}}.\end{aligned}$$

(4) *If* $m_1 = 0$, $m_3 = 0$, *then*

$$\begin{aligned}\Sigma &= \varrho_1^2 J_1^{-1}\Sigma_0\left(J_1^{-1}\right)^{\mathrm{T}} + \varrho_2^2\left(J_3J_2\right)^{-1}\Sigma_0\left[\left(J_3J_2\right)^{-1}\right]^{\mathrm{T}} \\ &\quad + \tilde{\varrho}_3^2\left(J_7J_5J_4\right)^{-1}\Theta_0\left[\left(J_7J_5J_4\right)^{-1}\right]^{\mathrm{T}} + \bar{\varrho}_4^2\left(Q_5Q_3Q_2Q_1\right)^{-1}\bar{\Theta}_0 \\ &\quad \left[\left(Q_5Q_3Q_2Q_1\right)^{-1}\right]^{\mathrm{T}}.\end{aligned}$$

(5) *If* $m_1 \neq 0$, $m_3 \neq 0$, *then*

$$\begin{aligned}\Sigma &= \varrho_1^2 J_1^{-1}\Sigma_0\left(J_1^{-1}\right)^{\mathrm{T}} + \varrho_2^2\left(J_3J_2\right)^{-1}\Sigma_0\left[\left(J_3J_2\right)^{-1}\right]^{\mathrm{T}} \\ &\quad + \varrho_3^2\left(J_6J_5J_4\right)^{-1}\Sigma_0\left[\left(J_6J_5J_4\right)^{-1}\right]^{\mathrm{T}} + \varrho_4^2\left(Q_4Q_3Q_2Q_1\right)^{-1}\Sigma_0 \\ &\quad \left[\left(Q_4Q_3Q_2Q_1\right)^{-1}\right]^{\mathrm{T}}.\end{aligned}$$

(6) *If* $m_1 \neq 0$, $m_3 = 0$, *then*

$$\begin{aligned}\Sigma &= \varrho_1^2 J_1^{-1}\Sigma_0\left(J_1^{-1}\right)^{\mathrm{T}} + \varrho_2^2\left(J_3J_2\right)^{-1}\Sigma_0\left[\left(J_3J_2\right)^{-1}\right]^{\mathrm{T}} \\ &\quad + \varrho_3^2\left(J_6J_5J_4\right)^{-1}\Sigma_0\left[\left(J_6J_5J_4\right)^{-1}\right]^{\mathrm{T}} + \bar{\varrho}_4^2\left(Q_5Q_3Q_2Q_1\right)^{-1}\bar{\Theta}_0 \\ &\quad \left[\left(Q_5Q_3Q_2Q_1\right)^{-1}\right]^{\mathrm{T}},\end{aligned}$$

with

$$J_1 = \begin{pmatrix} a_{21}a_{32}a_{43} & -a_{32}a_{43}(a_{21}+a_{32}+a_{43}) & a_{43}\left(a_{32}^2+a_{32}a_{43}+a_{43}^2\right) & a_{43}\left(a_{21}a_{32}-a_{43}^2\right) \\ 0 & a_{32}a_{43} & -a_{43}(a_{32}+a_{43}) & a_{43}^2 \\ 0 & 0 & a_{43} & -a_{43} \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$J_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$J_3 = \begin{pmatrix} -a_{14}a_{32}a_{43} & a_{14}a_{43}(a_{11}+a_{32}+a_{43}) & -a_{14}\left(a_{11}^2+a_{11}a_{43}+a_{43}^2\right) & -a_{11}^3 \\ 0 & -a_{14}a_{43} & a_{14}(a_{11}+a_{43}) & a_{11}^2 \\ 0 & 0 & -a_{14} & -a_{11} \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$J_4 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

$$J_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{a_{21}}{a_{14}} & 1 \end{pmatrix}, \quad J_7 = \begin{pmatrix} -a_{14}a_{43} & a_{14}(a_{11}+a_{43}) & a_{11}^2 & 0 \\ 0 & -a_{14} & -a_{11} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$J_6 = \begin{pmatrix} -a_{14}a_{43}m_1 & a_{14}(a_{11}+a_{21}+a_{43})m_1 & \left(a_{11}^2+a_{11}a_{21}+a_{21}^2\right)m_1 & -a_{21}^3 \\ 0 & -a_{14}m_1 & -(a_{11}+a_{21})m_1 & a_{21}^2 \\ 0 & 0 & m_1 & -a_{21} \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$Q_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad Q_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{a_{14}}{a_{21}} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad Q_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{a_{32}}{m_2} & 1 \end{pmatrix},$$

$$Q_4 = \begin{pmatrix} a_{21}m_2m_3 & -(a_{11}+a_{21}+a_{32})m_2m_3 & \left(a_{11}^2+a_{11}a_{32}+a_{32}^2-a_{11}a_{14}-a_{14}a_{21}-a_{14}a_{32}\right)m_3 & -a_{32}^3 \\ 0 & m_2m_3 & (a_{14}-a_{11}-a_{32})m_3 & a_{32}^2 \\ 0 & 0 & m_3 & -a_{32} \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$Q_5 = \begin{pmatrix} a_{21}m_2 & -(a_{11}+a_{21})m_2 & a_{11}^2-a_{11}a_{14}-a_{14}a_{21} & 0 \\ 0 & m_2 & a_{14}-a_{11} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad Q_6 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

$$Q_7 = \begin{pmatrix} a_{21}a_{32} & -a_{32}(a_{11}+a_{32}) & a_{32}^2 & a_{21}a_{32} \\ 0 & a_{32} & -a_{32} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\Sigma_0 = \begin{pmatrix} \frac{\tau_2\tau_3-\tau_1\tau_4}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & -\frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 \\ 0 & \frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} \\ -\frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & \frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 \\ 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & \frac{\tau_1\tau_2-\tau_3}{2\tau_4(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} \end{pmatrix},$$

$$\Theta_0 = \begin{pmatrix} \frac{l_2}{2(l_1 l_2 - l_3)} & 0 & -\frac{1}{2(l_1 l_2 - l_3)} & 0 \\ 0 & \frac{1}{2(l_1 l_2 - l_3)} & 0 & 0 \\ -\frac{1}{2(l_1 l_2 - l_3)} & 0 & \frac{l_1}{2l_3(l_1 l_2 - l_3)} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$\bar{\Theta}_0 = \begin{pmatrix} \frac{\bar{l}_2}{2(\bar{l}_1 \bar{l}_2 - \bar{l}_3)} & 0 & -\frac{1}{2(\bar{l}_1 \bar{l}_2 - \bar{l}_3)} & 0 \\ 0 & \frac{1}{2(\bar{l}_1 \bar{l}_2 - \bar{l}_3)} & 0 & 0 \\ -\frac{1}{2(\bar{l}_1 \bar{l}_2 - \bar{l}_3)} & 0 & \frac{\bar{l}_1}{2\bar{l}_3(\bar{l}_1 \bar{l}_2 - \bar{l}_3)} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$m_1 = \frac{a_{21}(a_{14} + a_{21} - a_{11})}{a_{14}},\ m_2 = \frac{a_{14}(a_{11} - a_{14} - a_{21})}{a_{21}},\ m_3 = \frac{a_{32}(a_{11} - a_{14} - a_{32})}{m_2},$$
$$\varrho_1 = a_{21}a_{32}a_{43}\sigma_1,$$
$$\varrho_2 = -a_{14}a_{32}a_{43}\sigma_2,\ \varrho_3 = -a_{14}a_{43}m_1\sigma_3,$$
$$\varrho_4 = a_{21}m_2m_3\sigma_4,\ \tilde{\varrho}_3 = -a_{14}a_{43}\sigma_3,\ \bar{\varrho}_4 = a_{21}m_2\sigma_4,\ \tilde{\varrho}_4 = a_{21}a_{32}\sigma_4,$$
$$\tau_1 = a_{11} + a_{21} + a_{32} + a_{43},\ \tau_2 = a_{11}a_{21} + a_{11}a_{32} + a_{11}a_{43} + a_{21}a_{32} + a_{21}a_{43} + a_{32}a_{43},$$
$$\tau_3 = a_{11}a_{21}a_{32} + a_{11}a_{21}a_{43} + a_{11}a_{32}a_{43},$$
$$\tau_4 = a_{14}a_{21}a_{32}a_{43},\ l_1 = a_{11} + a_{32} + a_{43},\ l_2 = a_{11}a_{32} + a_{11}a_{43} + a_{32}a_{43},$$
$$l_3 = a_{14}a_{32}a_{43},\ \bar{l}_1 = a_{11} + a_{21} + a_{43},$$
$$\bar{l}_2 = a_{11}a_{21} + a_{11}a_{43} + a_{21}a_{43},\ \bar{l}_3 = a_{14}a_{21}a_{43}.$$

Proof For convenience and simplicity, denote $B(t) = (B_1(t), B_2(t), B_3(t), B_4(t))^{\mathrm{T}}$ and

$$C = \begin{pmatrix} -a_{11} & 0 & 0 & -a_{14} \\ a_{21} & -a_{21} & 0 & a_{21} \\ 0 & a_{32} & -a_{32} & 0 \\ 0 & 0 & a_{43} & -a_{43} \end{pmatrix},\ H = \begin{pmatrix} \sigma_1 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 \\ 0 & 0 & \sigma_3 & 0 \\ 0 & 0 & 0 & \sigma_4 \end{pmatrix}.$$

With these notations, system (4.3) can be simplified to the following equivalent form

$$dX(t) = CX(t)dt + HdB(t).$$

In the light of the continuous Markov processes theory (Gardiner 1983), the probability density $\Phi(X)$ around the quasi-stable equilibrium P^* satisfies the Fokker–Planck

equation

$$\begin{aligned}&\frac{\partial}{\partial x_1}[(-a_{11}x_1 - a_{14}x_4)\Phi] + \frac{\partial}{\partial x_2}[(a_{21}x_1 - a_{21}x_2 + a_{21}x_4)\Phi]\\&+ \frac{\partial}{\partial x_3}[(a_{32}x_2 - a_{32}x_3)\Phi] + \frac{\partial}{\partial x_4}[(a_{43}x_3 - a_{43}x_4)\Phi]\\&- \sum_{i=1}^{4} \frac{\sigma_i^2}{2}\frac{\partial^2\Phi}{\partial x_i^2} = 0.\end{aligned} \tag{4.4}$$

Since the diffusion matrix H is a constant matrix, then the probability density $\Phi(X)$ can be described by a Gaussian distribution by the study of Roozen (1989), i.e.,

$$\Phi(X) = c\exp\left\{-\frac{1}{2}XQX^{\mathrm{T}}\right\},$$

where Q is a real symmetric matrix and c is a positive constant satisfying the normalization condition $\int_{\mathbb{R}^4}\Phi(X)\mathrm{d}X = 1$.

Substituting these results into (4.4), we can get the constant $c = (2\pi)^{-2}|\Sigma|^{-\frac{1}{2}}$ and Q satisfies the following algebraic equation

$$QH^2Q + QC + C^{\mathrm{T}}Q = 0. \tag{4.5}$$

If the matrix Q is positive definite, so it is invertible, we define $Q^{-1} = \Sigma$, then the algebraic equation (4.5) can be equivalently rewritten as

$$H^2 + C\Sigma + \Sigma C^{\mathrm{T}} = 0. \tag{4.6}$$

According to the finite independent superposition principle (Gardiner 1983), (4.6) is equivalent to the sum of the following four algebraic sub-equations,

$$H_i^2 + C\Sigma_i + \Sigma_i C^{\mathrm{T}} = 0,\ i = 1, 2, 3, 4,$$

where $H_1 = \mathrm{diag}(\sigma_1, 0, 0, 0)$, $H_2 = \mathrm{diag}(0, \sigma_2, 0, 0)$, $H_3 = \mathrm{diag}(0, 0, \sigma_3, 0)$, $H_4 = \mathrm{diag}(0, 0, 0, \sigma_4)$, and the symmetric matrices Σ_i $(i = 1, 2, 3, 4)$ are their solutions, respectively. It is easy to see that $\Sigma = \Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4$ and $H^2 = H_1^2 + H_2^2 + H_3^2 + H_4^2$.

In order to study the positive definiteness of Σ, we define the characteristic equation of matrix C as

$$\varphi_C(\lambda) = \lambda^4 + \tau_1\lambda^3 + \tau_2\lambda^2 + \tau_3\lambda + \tau_4 = 0,\ \lambda \text{ denotes the eigenvalue,} \tag{4.7}$$

where

$$\tau_1 = a_{11} + a_{21} + a_{32} + a_{43},$$

$$\tau_2 = a_{11}a_{21} + a_{11}a_{32} + a_{11}a_{43} + a_{21}a_{32} + a_{21}a_{43} + a_{32}a_{43},$$
$$\tau_3 = a_{11}a_{21}a_{32} + a_{11}a_{21}a_{43} + a_{11}a_{32}a_{43},$$
$$\tau_4 = a_{14}a_{21}a_{32}a_{43}.$$

In view of the expressions of H^*, I^*, D^*, V^* and $R_0^S > 1$, we can verify that

$$\tau_1 > 0,\ \tau_3 > 0,\ \tau_4 > 0,\ \tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4 > 0,$$

which implies that all the roots of the characteristic Eq. (4.7) have negative real-parts and so the matrix C is a Hurwitz matrix.

Since the matrices C and C_1 [see (4.9) below] are similar, in view of the similarity invariant of the characteristic polynomial in linear algebra, we obtain that $\varphi_C(\lambda)$ is the similarity invariant of matrix C. It is clear that τ_1, τ_2, τ_3 and τ_4 are also similarity invariants. Hence, there exists a unique standard R_1 matrix of C.

Now we are in the position to give the specific form of Σ and show its positive definiteness. We divide the proof process into four steps: (i) prove the matrix Σ_1 is positive definite in Step 1; (ii) prove the matrix Σ_2 is positive definite in Step 2 and the proof process is similar to Step 1. (iii) verify the matrix Σ_3 is at least semi-positive definite in Step 3. (iv) show the matrix Σ_4 is at least semi-positive definite in Step 4.

Step 1 Consider the algebraic equation

$$H_1^2 + C\Sigma_1 + \Sigma_1 C^{\mathrm{T}} = 0. \tag{4.8}$$

Let $C_1 = J_1 C J_1^{-1}$, where the standardized transformation matrix J_1 is given by

$$J_1 = \begin{pmatrix} a_{21}a_{32}a_{43} & -a_{32}a_{43}(a_{21}+a_{32}+a_{43}) & a_{43}(a_{32}^2 + a_{32}a_{43} + a_{43}^2) & a_{43}(a_{21}a_{32} - a_{43}^2) \\ 0 & a_{32}a_{43} & -a_{43}(a_{32}+a_{43}) & a_{43}^2 \\ 0 & 0 & a_{43} & -a_{43} \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Direct calculation leads to that

$$C_1 = C_0 = \begin{pmatrix} -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \tag{4.9}$$

where τ_1, τ_2, τ_3 and τ_4 are the same as in Lemma 4.1. Moreover, Eq. (4.8) can be transformed into the following form

$$J_1 H_1^2 J_1^{\mathrm{T}} + J_1 C J_1^{-1} J_1 \Sigma_1 J_1^{\mathrm{T}} + J_1 \Sigma_1 J_1^{\mathrm{T}} \left(J_1 C J_1^{-1}\right)^{\mathrm{T}} = 0.$$

We rewrite the above equation as the following form

$$H_0^2 + C_1\Sigma_0 + \Sigma_0 C_1^{\mathrm{T}} = 0,$$

where $\Sigma_0 = \varrho_1^{-2} J_1 \Sigma_1 J_1^{\mathrm{T}}$ and $\varrho_1 = a_{21}a_{32}a_{43}\sigma_1$. It is noticed that the matrix C has all negative real-part eigenvalues, then C_1 is a standard R_1 matrix. According to Lemma 4.1, we obtain that the matrix Σ_0 is positive definite whose specific form is as follows:

$$\Sigma_0 = \begin{pmatrix} \frac{\tau_2\tau_3 - \tau_1\tau_4}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & -\frac{\tau_3}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 \\ 0 & \frac{\tau_3}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} \\ -\frac{\tau_3}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & \frac{\tau_1}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 \\ 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} & 0 & \frac{\tau_1\tau_2 - \tau_3}{2\tau_4(\tau_1\tau_2\tau_3 - \tau_3^2 - \tau_1^2\tau_4)} \end{pmatrix}.$$

Thus, the matrix $\Sigma_1 = \varrho_1^2 J_1^{-1} \Sigma_0 (J_1^{-1})^{\mathrm{T}}$ is also positive definite.

Step 2 Consider the algebraic equation

$$H_2^2 + C\Sigma_2 + \Sigma_2 C^{\mathrm{T}} = 0. \tag{4.10}$$

Define $C_2 = J_2 C J_2^{-1}$, where

$$J_2 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

then we obtain

$$C_2 = \begin{pmatrix} -a_{21} & 0 & a_{21} & a_{21} \\ a_{32} & -a_{32} & 0 & 0 \\ 0 & a_{43} & -a_{43} & 0 \\ 0 & 0 & -a_{14} & -a_{11} \end{pmatrix}.$$

Similar to Step 1, define $C_3 = J_3 C_2 J_3^{-1}$, where the standardized transformation matrix J_3 takes the form

$$J_3 = \begin{pmatrix} -a_{14}a_{32}a_{43} & a_{14}a_{43}(a_{11} + a_{32} + a_{43}) & -a_{14}(a_{11}^2 + a_{11}a_{43} + a_{43}^2) & -a_{11}^3 \\ 0 & -a_{14}a_{43} & a_{14}(a_{11} + a_{43}) & a_{11}^2 \\ 0 & 0 & -a_{14} & -a_{11} \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

In view of the uniqueness of standard R_1 matrix, we can conclude that

$$C_3 = C_0 = \begin{pmatrix} -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

By letting $\Sigma_0 = \varrho_2^{-2}(J_3J_2)\Sigma_2(J_3J_2)^{\mathrm{T}}$, where $\varrho_2 = -a_{14}a_{32}a_{43}\sigma_2$, then Equation (4.10) is equivalent to the following form

$$H_0^2 + C_3\Sigma_0 + \Sigma_0 C_3^{\mathrm{T}} = 0.$$

By Lemma 4.1 and the result of C having all negative real-part eigenvalues again, we can obtain that Σ_0 is positive definite, which takes the form

$$\Sigma_0 = \begin{pmatrix} \frac{\tau_2\tau_3-\tau_1\tau_4}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & -\frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 \\ 0 & \frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} \\ -\frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & \frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 \\ 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & \frac{\tau_1\tau_2-\tau_3}{2\tau_4(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} \end{pmatrix}.$$

Therefore, the matrix $\Sigma_2 = \varrho_2^2(J_3J_2)^{-1}\Sigma_0[(J_3J_2)^{-1}]^{\mathrm{T}}$ is also positive definite.

Step 3 Consider the algebraic equation

$$H_3^2 + C\Sigma_3 + \Sigma_3 C^{\mathrm{T}} = 0. \tag{4.11}$$

Define $C_4 = J_5J_4C(J_5J_4)^{-1}$, where

$$J_4 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad J_5 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{a_{21}}{a_{14}} & 1 \end{pmatrix},$$

then we have

$$C_4 = \begin{pmatrix} -a_{32} & 0 & -\frac{a_{21}a_{32}}{a_{14}} & a_{32} \\ a_{43} & -a_{43} & 0 & 0 \\ 0 & -a_{14} & -a_{11} & 0 \\ 0 & 0 & m_1 & -a_{21} \end{pmatrix},$$

where

$$m_1 = \frac{a_{21}(a_{14}+a_{21}-a_{11})}{a_{14}}.$$

Based on the value of m_1, we will consider the following two cases:
(1) $m_1 \neq 0$; (2) $m_1 = 0$.

Case 1. If $m_1 \neq 0$, following the derivation process in Step 1, we define $C_5 = J_6 C_4 J_6^{-1}$, where the standardized transformation matrix J_6 is given by

$$J_6 = \begin{pmatrix} -a_{14}a_{43}m_1 & a_{14}(a_{11}+a_{21}+a_{43})m_1 & \left(a_{11}^2+a_{11}a_{21}+a_{21}^2\right)m_1 & -a_{21}^3 \\ 0 & -a_{14}m_1 & -(a_{11}+a_{21})m_1 & a_{21}^2 \\ 0 & 0 & m_1 & -a_{21} \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

By the uniqueness of the standard R_1 matrix, we can obtain that

$$C_5 = C_1 = \begin{pmatrix} -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Thus, Eq. (4.11) can be transformed into the following form

$$\begin{aligned} &(J_6J_5J_4)H_3^2(J_6J_5J_4)^{\mathrm{T}} + (J_6J_5J_4)C(J_6J_5J_4)^{-1}(J_6J_5J_4) \\ &\quad \Sigma_3(J_6J_5J_4)^{\mathrm{T}} + (J_6J_5J_4)\Sigma_3(J_6J_5J_4)^{\mathrm{T}}[(J_6J_5J_4)C(J_6J_5J_4)^{-1}]^T = 0, \end{aligned}$$

i.e.,

$$H_0^2 + C_5\Sigma_0 + \Sigma_0 C_5^{\mathrm{T}} = 0,$$

where $\Sigma_0 = \varrho_3^{-2}(J_6J_5J_4)\Sigma_3(J_6J_5J_4)^{\mathrm{T}}$ and $\varrho_3 = -a_{14}a_{43}m_1\sigma_3$. Noting that the matrix C has all negative real-part eigenvalues, then C_5 is a standard R_1 matrix. In view of Lemma 4.1, we can get that the matrix Σ_0 is positive definite and the specific form of Σ_0 is as follows:

$$\Sigma_0 = \begin{pmatrix} \frac{\tau_2\tau_3-\tau_1\tau_4}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & -\frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 \\ 0 & \frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} \\ -\frac{\tau_3}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & \frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 \\ 0 & -\frac{\tau_1}{2(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} & 0 & \frac{\tau_1\tau_2-\tau_3}{2\tau_4(\tau_1\tau_2\tau_3-\tau_3^2-\tau_1^2\tau_4)} \end{pmatrix}.$$

Hence, the matrix $\Sigma_3 = \varrho_3^2(J_6J_5J_4)^{-1}\Sigma_0[(J_6J_5J_4)^{-1}]^{\mathrm{T}}$ is also positive definite.

Case 2. If $m_1 = 0$, i.e., $a_{11} - a_{21} = a_{14}$, define $C_6 = J_7 C_4 J_7^{-1}$, where the standardized transformation matrix J_7 takes the form

$$J_7 = \begin{pmatrix} -a_{14}a_{43} & a_{14}(a_{11}+a_{43}) & a_{11}^2 & 0 \\ 0 & -a_{14} & -a_{11} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

By a simple computation, we have

$$C_6 = \begin{pmatrix} -l_1 & -l_2 & -l_3 & -l_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -a_{21} \end{pmatrix},$$

where

$$\begin{aligned} &l_1 = a_{11} + a_{32} + a_{43} > 0,\ l_2 = a_{11}a_{32} + a_{11}a_{43} + a_{32}a_{43} > 0, \\ &l_3 = a_{14}a_{32}a_{43} > 0,\ l_4 = a_{14}a_{32}a_{43} > 0. \end{aligned} \tag{4.12}$$

By (4.12) and the expressions of a_{21}, a_{32} and a_{43}, we can easily validate that $l_1 l_2 - l_3 > (a_{11} - a_{14})a_{32}a_{43} = a_{21}a_{32}a_{43} > 0$. That is to say, the conditions of Lemma 4.2 are satisfied, hence C_6 is a standard R_2 matrix and the corresponding matrix Θ_0 is semi-positive definite. Similarly, Eq. (4.11) can be transformed into the following form

$$\begin{aligned} &(J_7J_5J_4)H_3^2(J_7J_5J_4)^{\mathrm{T}} + (J_7J_5J_4)C(J_7J_5J_4)^{-1}(J_7J_5J_4) \\ &\quad \Sigma_3(J_7J_5J_4)^{\mathrm{T}} + (J_7J_5J_4)\Sigma_3(J_7J_5J_4)^{\mathrm{T}}[(J_7J_5J_4)C(J_7J_5J_4)^{-1}]^T = 0, \end{aligned}$$

which can be reduced to

$$H_0^2 + C_6\Theta_0 + \Theta_0 C_6^{\mathrm{T}} = 0,$$

where $\Theta_0 = \tilde{\varrho}_3^{-2}(J_7J_5J_4)\Sigma_3(J_7J_5J_4)^{\mathrm{T}}$ and $\tilde{\varrho}_3 = -a_{14}a_{43}\sigma_3$. From Lemma 4.2, it follows that the specific expression of the semi-positive definite matrix Θ_0 is as follows:

$$\Theta_0 = \begin{pmatrix} \dfrac{l_2}{2(l_1l_2 - l_3)} & 0 & -\dfrac{1}{2(l_1l_2 - l_3)} & 0 \\ 0 & \dfrac{1}{2(l_1l_2 - l_3)} & 0 & 0 \\ -\dfrac{1}{2(l_1l_2 - l_3)} & 0 & \dfrac{l_1}{2l_3(l_1l_2 - l_3)} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Consequently, the matrix $\Sigma_3 = \tilde{\varrho}_3^2(J_7J_5J_4)^{-1}\Theta_0[(J_7J_5J_4)^{-1}]^{\mathrm{T}}$ is also semi-positive definite.

Step 4. Consider the algebraic equation

$$H_4^2 + C\Sigma_4 + \Sigma_4 C^{\mathrm{T}} = 0. \tag{4.13}$$

Let $P_1 = Q_1 C Q_1^{-1}$, where

$$Q_1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad P_1 = \begin{pmatrix} -a_{43} & 0 & 0 & a_{43} \\ a_{21} & -a_{21} & a_{21} & 0 \\ -a_{14} & 0 & -a_{11} & 0 \\ 0 & a_{32} & 0 & -a_{32} \end{pmatrix}.$$

Then, we define $P_2 = Q_2 P_1 Q_2^{-1}$, where the elimination matrix Q_2 takes the form

$$Q_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \frac{a_{14}}{a_{21}} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Direct calculation leads to that

$$P_2 = \begin{pmatrix} -a_{43} & 0 & 0 & a_{43} \\ a_{21} & -a_{14} - a_{21} & a_{21} & 0 \\ 0 & m_2 & a_{14} - a_{11} & 0 \\ 0 & a_{32} & 0 & -a_{32} \end{pmatrix},$$

where

$$m_2 = \frac{a_{14}(a_{11} - a_{14} - a_{21})}{a_{21}}.$$

According to the value of m_2, we will discuss it in two cases:
(1) $m_2 \neq 0$; (2) $m_2 = 0$.

Case 1. If $m_2 \neq 0$, define $P_3 = Q_3 P_2 Q_3^{-1}$, where the elimination matrix Q_3 is given by

$$Q_3 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{a_{32}}{m_2} & 1 \end{pmatrix}.$$

By a simple computation, we obtain

$$P_3 = \begin{pmatrix} -a_{43} & 0 & \frac{a_{32}a_{43}}{m_2} & a_{43} \\ a_{21} & -a_{14} - a_{21} & a_{21} & 0 \\ 0 & m_2 & a_{14} - a_{11} & 0 \\ 0 & 0 & m_3 & -a_{32} \end{pmatrix},$$

where

$$m_3 = \frac{a_{32}(a_{11} - a_{14} - a_{32})}{m_2}.$$

In view of the value of m_3, we will consider the following two subcases:
(i) $m_3 \neq 0$; (ii) $m_3 = 0$.

Case 1.1. If $m_3 \neq 0$, following the derivation process in Step 1, we define $P_4 = Q_4 P_3 Q_4^{-1}$, where the standardized transformation matrix Q_4 is given by

$$Q_4 = \begin{pmatrix} a_{21}m_2m_3 & -(a_{11}+a_{21}+a_{32})m_2m_3 & (a_{11}^2 + a_{11}a_{32} + a_{32}^2 - a_{11}a_{14} - a_{14}a_{21} - a_{14}a_{32})m_3 & -a_{32}^3 \\ 0 & m_2m_3 & (a_{14} - a_{11} - a_{32})m_3 & a_{32}^2 \\ 0 & 0 & m_3 & -a_{32} \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

In the light of the uniqueness of the standard R_1 matrix, we can calculate that

$$P_4 = C_0 = \begin{pmatrix} -\tau_1 & -\tau_2 & -\tau_3 & -\tau_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Hence, Eq. (4.13) can be transformed into the following form

$$\begin{aligned} &(Q_4Q_3Q_2Q_1)H_4^2(Q_4Q_3Q_2Q_1)^{\mathrm{T}} \\ &+ (Q_4Q_3Q_2Q_1)C(Q_4Q_3Q_2Q_1)^{-1}(Q_4Q_3Q_2Q_1)\Sigma_4(Q_4Q_3Q_2Q_1)^{\mathrm{T}} \\ &+ (Q_4Q_3Q_2Q_1)\Sigma_4(Q_4Q_3Q_2Q_1)^{\mathrm{T}}[(Q_4Q_3Q_2Q_1)C(Q_4Q_3Q_2Q_1)^{-1}]^{\mathrm{T}} = 0, \end{aligned}$$

i.e.,

$$H_0^2 + P_4\Sigma_0 + \Sigma_0 P_4^{\mathrm{T}} = 0,$$

where $\Sigma_0 = \varrho_4^{-2}(Q_4Q_3Q_2Q_1)\Sigma_4(Q_4Q_3Q_2Q_1)^{\mathrm{T}}$ and $\varrho_4 = a_{21}m_2m_3\sigma_4$. According to Lemma 4.1, we obtain that the matrix Σ_0 is positive definite. Accordingly, the matrix $\Sigma_4 = \varrho_4^2(Q_4Q_3Q_2Q_1)^{-1}\Sigma_0[(Q_4Q_3Q_2Q_1)^{-1}]^{\mathrm{T}}$ is also positive definite.

Case 1.2. If $m_3 = 0$, i.e., $a_{11} - a_{32} = a_{14}$, define $P_5 = Q_5 P_3 Q_5^{-1}$, where the standardized transformation matrix Q_5 is denoted by

$$Q_5 = \begin{pmatrix} a_{21}m_2 & -(a_{11}+a_{21})m_2 & a_{11}^2 - a_{11}a_{14} - a_{14}a_{21} & 0 \\ 0 & m_2 & a_{14} - a_{11} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Direct calculation yields that

$$P_5 = \begin{pmatrix} -\bar{l}_1 & -\bar{l}_2 & -\bar{l}_3 & -\bar{l}_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -a_{32} \end{pmatrix},$$

where

$$\begin{aligned} &\bar{l}_1 = a_{11} + a_{21} + a_{43} > 0, \quad \bar{l}_2 = a_{11}a_{21} + a_{11}a_{43} + a_{21}a_{43} > 0, \\ &\bar{l}_3 = a_{14}a_{21}a_{43} > 0, \quad \bar{l}_4 = a_{14}a_{43}(a_{11} - a_{14} - a_{21}). \end{aligned} \tag{4.14}$$

In view of (4.14) and the expressions of a_{21}, a_{32} and a_{43}, we can easily conclude that $\bar{l}_1\bar{l}_2 - \bar{l}_3 > (a_{11} - a_{14})a_{21}a_{43} = a_{21}a_{32}a_{43} > 0$. In other words, the conditions of Lemma 4.2 hold, so P_5 is a standard R_2 matrix and the corresponding matrix $\bar{\Theta}_0$ is semi-positive definite. Similarly, Eq. (4.13) can be transformed into the following form

$$\begin{aligned} &(Q_5Q_3Q_2Q_1)H_4^2(Q_5Q_3Q_2Q_1)^{\mathrm{T}} + (Q_5Q_3Q_2Q_1)C(Q_5Q_3Q_2Q_1)^{-1} \\ &\quad (Q_5Q_3Q_2Q_1)\Sigma_4(Q_5Q_3Q_2Q_1)^{\mathrm{T}} \\ &\quad + (Q_5Q_3Q_2Q_1)\Sigma_4(Q_5Q_3Q_2Q_1)^{\mathrm{T}}[(Q_5Q_3Q_2Q_1)C(Q_5Q_3Q_2Q_1)^{-1}]^{\mathrm{T}} = 0, \end{aligned}$$

which is equivalent to

$$H_0^2 + P_5\bar{\Theta}_0 + \bar{\Theta}_0P_5^{\mathrm{T}} = 0,$$

where $\bar{\Theta}_0 = \bar{\varrho}_4^{-2}(Q_5Q_3Q_2Q_1)\Sigma_4(Q_5Q_3Q_2Q_1)^{\mathrm{T}}$ and $\bar{\varrho}_4 = a_{21}m_2\sigma_4$. By Lemma 4.2, we obtain that the specific expression of the semi-positive definite matrix $\bar{\Theta}_0$ takes the form

$$\bar{\Theta}_0 = \begin{pmatrix} \dfrac{\bar{l}_2}{2(\bar{l}_1\bar{l}_2 - \bar{l}_3)} & 0 & -\dfrac{1}{2(\bar{l}_1\bar{l}_2 - \bar{l}_3)} & 0 \\ 0 & \dfrac{1}{2(\bar{l}_1\bar{l}_2 - \bar{l}_3)} & 0 & 0 \\ -\dfrac{1}{2(\bar{l}_1\bar{l}_2 - \bar{l}_3)} & 0 & \dfrac{\bar{l}_1}{2\bar{l}_3(\bar{l}_1\bar{l}_2 - \bar{l}_3)} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Hence, the matrix $\Sigma_4 = \bar{\varrho}_4^2(Q_5Q_3Q_2Q_1)^{-1}\bar{\Theta}_0[(Q_5Q_3Q_2Q_1)^{-1}]^{\mathrm{T}}$ is also semi-positive definite.

Case 2. If $m_2 = 0$, i.e., $a_{11} - a_{21} = a_{14}$, define $P_6 = Q_6P_2Q_6^{-1}$, where

$$Q_6 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

then we have

$$P_6 = \begin{pmatrix} -a_{43} & 0 & a_{43} & 0 \\ a_{21} & -a_{14} - a_{21} & 0 & a_{21} \\ 0 & a_{32} & -a_{32} & 0 \\ 0 & 0 & 0 & a_{14} - a_{11} \end{pmatrix} = \begin{pmatrix} -a_{43} & 0 & a_{43} & 0 \\ a_{21} & -a_{11} & 0 & a_{21} \\ 0 & a_{32} & -a_{32} & 0 \\ 0 & 0 & 0 & -a_{21} \end{pmatrix}.$$

Next, we define $P_7 = Q_7 P_6 Q_7^{-1}$, where the standardized transformation matrix Q_7 takes the form

$$Q_7 = \begin{pmatrix} a_{21}a_{32} & -a_{32}(a_{11} + a_{32}) & a_{32}^2 & a_{21}a_{32} \\ 0 & a_{32} & -a_{32} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

By a simple computation, we have

$$P_7 = \begin{pmatrix} -\tilde{l}_1 & -\tilde{l}_2 & -\tilde{l}_3 & -\tilde{l}_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -a_{21} \end{pmatrix},$$

where

$$\begin{aligned} &\tilde{l}_1 = l_1 = a_{11} + a_{32} + a_{43} > 0,\ \tilde{l}_2 = l_2 = a_{11}a_{32} + a_{11}a_{43} + a_{32}a_{43} > 0, \\ &\tilde{l}_3 = l_3 = a_{14}a_{32}a_{43} > 0,\ \tilde{l}_4 = a_{21}a_{32}(a_{21} - a_{43}). \end{aligned}$$

Since the conditions of Lemma 4.2 hold, thus P_7 is a standard R_2 matrix and the corresponding matrix Θ_0 is semi-positive definite. Similarly, Eq. (4.13) can be transformed into the following form

$$\begin{aligned} &(Q_7Q_6Q_2Q_1)H_4^2(Q_7Q_6Q_2Q_1)^{\mathrm{T}} \\ &\quad + (Q_7Q_6Q_2Q_1)C(Q_7Q_6Q_2Q_1)^{-1}(Q_7Q_6Q_2Q_1)\Sigma_4(Q_7Q_6Q_2Q_1)^{\mathrm{T}} \\ &\quad + (Q_7Q_6Q_2Q_1)\Sigma_4(Q_7Q_6Q_2Q_1)^{\mathrm{T}}[(Q_7Q_6Q_2Q_1)C(Q_7Q_6Q_2Q_1)^{-1}]^{\mathrm{T}} = 0, \end{aligned}$$

that is,

$$H_0^2 + P_7\Theta_0 + \Theta_0 P_7^{\mathrm{T}} = 0,$$

where $\Theta_0 = \tilde{\varrho}_4^{-2}(Q_7Q_6Q_2Q_1)\Sigma_4(Q_7Q_6Q_2Q_1)^{\mathrm{T}}$ and $\tilde{\varrho}_4 = a_{21}a_{32}\sigma_4$. In view of Lemma 4.2, we obtain that the specific form of the semi-positive definite matrix Θ_0 is as follows:

$$\Theta_0 = \begin{pmatrix} \frac{l_2}{2(l_1l_2-l_3)} & 0 & -\frac{1}{2(l_1l_2-l_3)} & 0 \\ 0 & \frac{1}{2(l_1l_2-l_3)} & 0 & 0 \\ -\frac{1}{2(l_1l_2-l_3)} & 0 & \frac{l_1}{2l_3(l_1l_2-l_3)} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Therefore, the matrix $\Sigma_4 = \tilde{\varrho}_4^2(Q_7Q_6Q_2Q_1)^{-1}\Theta_0[(Q_7Q_6Q_2Q_1)^{-1}]^{\mathrm{T}}$ is also semi-positive definite.

In summary, if $R_0^S > 1$ holds, then the matrices Σ_1 and Σ_2 are both positive definite, and the matrices Σ_3 and Σ_4 are at least semi-positive definite. Thus, the matrix $\Sigma = \Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4$ in Eq. (4.6) is positive definite. Finally, by the relationship between systems (4.1) and (4.3), we obtain that the stationary distribution $\pi(\cdot)$ around P^* follows a unique log-normal probability density $\Phi(H, I, D, V)$, which takes the form

$$\Phi(H, I, D, V) = (2\pi)^{-2}|\Sigma|^{-\frac{1}{2}}e^{-\frac{1}{2}(\ln\frac{H}{H^*},\ln\frac{I}{I^*},\ln\frac{D}{D^*},\ln\frac{V}{V^*})\Sigma^{-1}(\ln\frac{H}{H^*},\ln\frac{I}{I^*},\ln\frac{D}{D^*},\ln\frac{V}{V^*})^{\mathrm{T}}},$$

where the specific form of Σ can be determined by the above discussion. This completes the proof. □

5 Extinction

In this section, we will establish sufficient criteria for extinction of the infected hepatocytes.

Theorem 5.1 *Let $(H(t), I(t), D(t), V(t))^{\mathrm{T}}$ be the solution of system (1.2 with any initial value $(H(0), I(0), D(0), V(0))^{\mathrm{T}} \in \mathbb{R}_+^4$. If $\mu > \sigma_1^2/2$, then for almost $\omega \in \Omega$, we have*

$$\limsup_{t\to\infty} \frac{1}{t}\ln\left(\frac{a}{\delta(\alpha+\beta)}I(t) + \frac{R_0^{\frac{1}{3}}}{\alpha+\beta}D(t) + \frac{R_0^{\frac{2}{3}}}{\beta}V(t)\right) \le \nu \text{ a.s.}$$

If $\nu < 0$, then the populations I, D and V will go to extinction exponentially a.s. and the distribution of $H(t)$ converges weakly to the invariant measure υ in $\mathbb{R}_+$ which has the density $f_*(x)$, where

$$\begin{aligned}\nu = &\min\{\delta, \alpha+\beta, c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\le 1\}} + \max\{\delta, \alpha+\beta, c\} \\ &\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}} - \big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1} \\ &+ \frac{a\beta ks\sigma_1}{\mu\delta R_0^{\frac{2}{3}}(\alpha+\beta)\sqrt{2\mu-\sigma_1^2}},\end{aligned}$$

$$R_0 = \frac{a\beta sk}{c\delta\mu(\alpha+\beta)},\ f_*(x) = \mathbb{C}x^{-2}x^{-\frac{2\mu}{\sigma_1^2}}e^{-\frac{2s}{\sigma_1^2 x}},\ x \in (0,\infty),$$
$$\mathbb{C} = \left[\sigma_1^{-2}\left(\frac{\sigma_1^2}{2s}\right)^{\frac{2\mu}{\sigma_1^2}+1}\Gamma\left(\frac{2\mu}{\sigma_1^2}+1\right)\right]^{-1}.$$

Proof Since for any initial value $(H(0), I(0), D(0), V(0))^{\mathrm{T}} \in \mathbb{R}_+^4$, the solution of system (1.2) is positive. Then, we have

$$dH(t) \le [s - \mu H(t)]dt + \sigma_1 H(t)dB_1(t).$$

Let $X(t)$ be the solution of the following stochastic differential equation

$$dX(t) = [s - \mu X(t)]dt + \sigma_1 X(t)dB_1(t),$$

with the initial value $X(0) = H(0) > 0$, then by the comparison theorem of one-dimensional stochastic differential equation (Peng and Zhu 2006), we obtain $H(t) \le X(t)$ for any $t \ge 0$ a.s.

Moreover, according to Theorem 3.1 in Liu et al. (2012), we can obtain that the process $X(t)$ converges weakly to an invariant distribution v in $\mathbb{R}_+$ which has the density $f_*(x)$, where

$$f_*(x) = \mathbb{C}x^{-2}x^{-\frac{2\mu}{\sigma_1^2}}e^{-\frac{2s}{\sigma_1^2 x}},\ x \in (0,\infty),$$

and

$$\mathbb{C} = \left[\sigma_1^{-2}\left(\frac{\sigma_1^2}{2s}\right)^{\frac{2\mu}{\sigma_1^2}+1}\Gamma\left(\frac{2\mu}{\sigma_1^2}+1\right)\right]^{-1}.$$

From the ergodic theorem, it gives that

$$\lim_{t\to\infty}\frac{1}{t}\int_0^{\mathrm{T}} X(\varpi)\mathrm{d}\varpi = \int_0^\infty x v(\mathrm{d}x) = H_0 = \frac{s}{\mu}\ \text{a.s..}$$

On the other hand, let

$$\sqrt[3]{R_0}(\omega_1,\omega_2,\omega_3) = (\omega_1,\omega_2,\omega_3)M_0,$$

where $(\omega_1,\omega_2,\omega_3) = (a/(\alpha+\beta), R_0^{\frac{1}{3}}, cR_0^{\frac{2}{3}}/\beta)$ and $M_0 = \begin{pmatrix} 0 & 0 & \frac{ks}{\mu\delta} \\ \frac{a}{\alpha+\beta} & 0 & 0 \\ 0 & \frac{\beta}{c} & 0 \end{pmatrix}$.

Define a C^2-function $\tilde{U} : \mathbb{R}_+^3 \to \mathbb{R}_+$ by

$$\tilde{U}(I, D, V) = \rho_1 I + \rho_2 D + \rho_3 V,$$

where $\rho_1 = \omega_1/\delta$, $\rho_2 = \omega_2/(\alpha+\beta)$, $\rho_3 = \omega_3/c$. Applying Itô's formula (Mao 1997) to $\ln \tilde{U}$, we obtain

$$d(\ln \tilde{U}) = L(\ln \tilde{U})dt + \frac{\rho_1\sigma_2 I}{\tilde{U}} dB_2(t) + \frac{\rho_2\sigma_3 D}{\tilde{U}} dB_3(t) + \frac{\rho_3\sigma_4 V}{\tilde{U}} dB_4(t), \tag{5.1}$$

where

$$\begin{aligned} L(\ln \tilde{U}) = {} & \frac{\rho_1}{\tilde{U}}[kHV - \delta I] + \frac{\rho_2}{\tilde{U}}[aI - (\alpha+\beta)D] + \frac{\rho_3}{\tilde{U}}[\beta D - cV] \\ & - \frac{\rho_1^2\sigma_2^2 I^2}{2\tilde{U}^2} - \frac{\rho_2^2\sigma_3^2 D^2}{2\tilde{U}^2} - \frac{\rho_3^2\sigma_4^2 V^2}{2\tilde{U}^2}. \end{aligned}$$

Moreover, we get

$$\begin{aligned} \tilde{U}^2 = {} & \left(\sigma_2\rho_1 I\frac{1}{\sigma_2} + \sigma_3\rho_2 D\frac{1}{\sigma_3} + \sigma_4\rho_3 V\frac{1}{\sigma_4}\right)^2 \le \left(\sigma_2^2\rho_1^2 I^2 + \sigma_3^2\rho_2^2 D^2 + \sigma_4^2\rho_3^2 V^2\right) \\ & \left(\frac{1}{\sigma_2^2} + \frac{1}{\sigma_3^2} + \frac{1}{\sigma_4^2}\right), \end{aligned} \tag{5.2}$$

and

$$
\begin{aligned}
&\frac{1}{\tilde{U}}\{\rho_1[kHV-\delta I]+\rho_2[aI-(\alpha+\beta)D]+\rho_3[\beta D-cV]\}\\
&\quad=\frac{\rho_1 V}{\tilde{U}}(kH-kH_0)+\frac{1}{\tilde{U}}\{\rho_1[kH_0V-\delta I]\\
&\qquad+\rho_2[aI-(\alpha+\beta)D]+\rho_3[\beta D-cV]\}\\
&\quad=\frac{\rho_1 kV(H-H_0)}{\tilde{U}}+\frac{1}{\tilde{U}}\left\{\frac{\omega_1}{\delta}[kH_0V-\delta I]+\frac{\omega_2}{\alpha+\beta}[aI-(\alpha+\beta)D]\right.\\
&\qquad\left.+\frac{\omega_3}{c}[\beta D-cV]\right\}\\
&\quad\le\frac{\rho_1 kV(X-H_0)}{\tilde{U}}+\frac{1}{\tilde{U}}\left\{\frac{\omega_1}{\delta}[kH_0V-\delta I]+\frac{\omega_2}{\alpha+\beta}[aI-(\alpha+\beta)D]\right.\\
&\qquad\left.+\frac{\omega_3}{c}[\beta D-cV]\right\}\\
&\quad\le\frac{\rho_1 k}{\rho_3}|X-H_0|+\frac{1}{\tilde{U}}(\omega_1,\omega_2,\omega_3)(M_0(I,D,V)^{\mathrm{T}}-(I,D,V)^{\mathrm{T}})\\
&\quad=\frac{\rho_1 k}{\rho_3}|X-H_0|+\frac{1}{\tilde{U}}(\sqrt[3]{R_0}-1)(\omega_1 I+\omega_2 D+\omega_3 V)\\
&\quad=\frac{\rho_1 k}{\rho_3}|X-H_0|+\frac{1}{\tilde{U}}(\sqrt[3]{R_0}-1)(\delta\rho_1 I+(\alpha+\beta)\rho_2 D+c\rho_3 V)\\
&\quad\le\min\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\le 1\}}+\max\{\delta,\alpha+\beta,c\}\\
&\qquad\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}}+\frac{\rho_1 k}{\rho_3}|X-H_0|.
\end{aligned}
\tag{5.3}
$$

According to (5.2) and (5.3), we have

$$
\begin{aligned}
L(\ln\tilde{U})\le{}&\min\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\le 1\}}+\max\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}}\\
&-\big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1}+\frac{\rho_1 k}{\rho_3}|X-H_0|.
\end{aligned}
$$

From (5.1) it follows that

$$
\begin{aligned}
d(\ln\tilde{U})\le{}&\Bigg(\min\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\le 1\}}+\max\{\delta,\alpha+\beta,c\}\\
&\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}}-\big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1}+\frac{\rho_1 k}{\rho_3}|X-H_0|\Bigg)dt\\
&+\frac{\rho_1\sigma_2 I}{\tilde{U}}dB_2(t)+\frac{\rho_2\sigma_3 D}{\tilde{U}}dB_3(t)+\frac{\rho_3\sigma_4 V}{\tilde{U}}dB_4(t).
\end{aligned}
\tag{5.4}
$$

Integrating (5.4) from 0 to t and then dividing by t on both sides leads to that

$$\begin{aligned}\frac{\ln \tilde{U}(t)}{t} \leq & \frac{\ln \tilde{U}(0)}{t} + \min\{\delta, \alpha+\beta, c\} \\ & \big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\leq 1\}} + \max\{\delta, \alpha+\beta, c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}} \\ & - \big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1} \\ & + \frac{\rho_1 k}{\rho_3 t}\int_0^t |X(u)-H_0|\mathrm{d}u + \frac{1}{t}\int_0^t \frac{\rho_1\sigma_2 I(u)}{\tilde{U}(u)}\mathrm{d}B_2(u) \\ & + \frac{1}{t}\int_0^t \frac{\rho_2\sigma_3 D(u)}{\tilde{U}(u)}\mathrm{d}B_3(u) + \frac{1}{t}\int_0^t \frac{\rho_3\sigma_4 V(u)}{\tilde{U}(u)}\mathrm{d}B_4(u) \\ = & \frac{\ln \tilde{U}(0)}{t} + \min\{\delta, \alpha+\beta, c\}(\sqrt[3]{R_0}-1)\mathbf{I}_{\{R_0\leq 1\}} \\ & + \max\{\delta, \alpha+\beta, c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}} - \big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1} \\ & + \frac{\rho_1 k}{\rho_3 t}\int_0^t |X(u)-H_0|\mathrm{d}u + \frac{M_1(t)}{t} + \frac{M_2(t)}{t} + \frac{M_3(t)}{t},\end{aligned} \tag{5.5}$$

where

$$M_1(t) := \int_0^t \frac{\rho_1\sigma_2 I(u)}{\tilde{U}(u)}\mathrm{d}B_2(u),\ M_2(t) := \int_0^t \frac{\rho_2\sigma_3 D(u)}{\tilde{U}(u)}\mathrm{d}B_3(u),$$
$$M_3(t) := \int_0^t \frac{\rho_3\sigma_4 V(u)}{\tilde{U}(u)}\mathrm{d}B_4(u)$$

are local martingales whose quadratic variations are

$$\begin{aligned}\langle\ M_1, M_1\rangle_t &= \sigma_2^2\int_0^t \left(\frac{\rho_1 I(u)}{\tilde{U}(u)}\right)^2 \mathrm{d}s \\ &\leq \sigma_2^2 t,\ \langle M_2, M_2\rangle_t = \sigma_3^2\int_0^t \left(\frac{\rho_2 D(u)}{\tilde{U}(u)}\right)^2 \mathrm{d}s \leq \sigma_3^2 t,\end{aligned}$$

$$\langle M_3, M_3\rangle_t = \sigma_4^2\int_0^t \left(\frac{\rho_3 V(u)}{\tilde{U}(u)}\right)^2 \mathrm{d}s \leq \sigma_4^2 t.$$

Employing the strong law of large numbers for local martingales (Mao 1997) yields that

$$\lim_{t\to\infty}\frac{M_i(t)}{t} = 0 \text{ a.s.},\ i = 1, 2, 3. \tag{5.6}$$

Since $H(t)$ is ergodic and $\int_0^\infty xv(\mathrm{d}x) < \infty$, we have

$$\begin{aligned}\lim_{t\to\infty}\frac{1}{t}\int_0^t |X(u)-H_0|\mathrm{d}u &= \int_0^\infty |x-H_0|v(\mathrm{d}x)\\ &\le \left(\int_0^\infty (x-H_0)^2 v(\mathrm{d}x)\right)^{\frac{1}{2}}\\ &= \left(\int_0^\infty x^2 v(\mathrm{d}x) - 2H_0\int_0^\infty xv(\mathrm{d}x) + H_0^2\right)^{\frac{1}{2}} \qquad (5.7)\\ &= \left(\frac{2s^2}{\mu(2\mu-\sigma_1^2)} - \frac{2s^2}{\mu^2} + \frac{s^2}{\mu^2}\right)^{\frac{1}{2}}\\ &= \frac{s\sigma_1}{\mu\sqrt{2\mu-\sigma_1^2}}.\end{aligned}$$

Taking the superior limit on both sides of (5.5) which together with (5.6) and (5.7) leads to that

$$\begin{aligned}&\limsup_{t\to\infty}\frac{\ln\tilde{U}(t)}{t}\\ &\le \min\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\le 1\}}\\ &\quad+\max\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}} - \big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1}\\ &\quad+\frac{\rho_1 ks\sigma_1}{\rho_3\mu\sqrt{2\mu-\sigma_1^2}}\\ &= \min\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\le 1\}}\\ &\quad+\max\{\delta,\alpha+\beta,c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}} - \big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1}\\ &\quad+\frac{a\beta ks\sigma_1}{\mu\delta R_0^{\frac{2}{3}}(\alpha+\beta)\sqrt{2\mu-\sigma_1^2}}\\ &:= \nu \text{ a.s.},\end{aligned}$$

which is the desired statement. In addition, if $\nu < 0$, we can conclude that

$$\lim_{t\to\infty}\frac{\ln I(t)}{t} = 0,\ \lim_{t\to\infty}\frac{\ln D(t)}{t} = 0,;\ \lim_{t\to\infty}\frac{\ln V(t)}{t} = 0;\ \text{a.s.},$$

which implies that $\lim_{t\to\infty} I(t) = 0$, $\lim_{t\to\infty} D(t) = 0$ and $\lim_{t\to\infty} V(t) = 0$ a.s.

Hence for any small $\epsilon > 0$, there exist t_0 and a set $\Omega_\epsilon \subset \Omega$ such that $\mathbb{P}(\Omega_\epsilon) > 1-\epsilon$ and $kHV \le \epsilon H$ for $t \ge t_0$ and $\omega \in \Omega_\epsilon$. Now from

$$\begin{aligned}&[s-\mu H(t)-\epsilon H(t)]dt + \sigma_1 H(t)dB_1(t)\\ &\le dH(t) \le [s-\mu H(t)]dt + \sigma_1 H(t)dB_1(t),\end{aligned}$$

Table 2 List of parameter values of system (1.1)

Parameters	Values	Source
s	2.6×10^7	Dahari et al. (2007)
k	1.67×10^{-12}	Murray et al. (2006)
μ	0.01	Dahari et al. (2007)
δ	0.053	Murray et al. (2006)
a	150	Murray et al. (2006)
β	0.87	Murray et al. (2006)
α	0.25	Estimated
c	3.8	Murray et al. (2006)

it follows that the distribution of the process $H(t)$ converges weakly to the invariant measure v in $\mathbb{R}_+$ which has the density $f_*(x)$. This completes the proof. □

6 Numerical Simulations

In this section, numerical simulations are carried out to verify the theoretical results. For the stochastic system (1.2), we adopt Milstein higher-order method developed in Higham (2001) and the discretization form of the stochastic system is given by:

$$\begin{cases} H_{j+1} = H_j + [s - kH_jV_j - \mu H_j]\Delta t + \sigma_1 H_j\sqrt{\Delta t}\varepsilon_{1,j} + \dfrac{\sigma_1^2}{2}H_j\big(\varepsilon_{1,j}^2 - 1\big)\Delta t, \\ I_{j+1} = I_j + [kH_jV_j - \delta I_j]\Delta t + \sigma_2 I_j\sqrt{\Delta t}\varepsilon_{2,j} + \dfrac{\sigma_2^2}{2}I_j\big(\varepsilon_{2,j}^2 - 1\big)\Delta t, \\ D_{j+1} = D_j + [aI_j - (\alpha+\beta)D_j]\Delta t + \sigma_3 D_j\sqrt{\Delta t}\varepsilon_{3,j} + \dfrac{\sigma_3^2}{2}D_j\big(\varepsilon_{3,j}^2 - 1\big)\Delta t, \\ V_{j+1} = V_j + [\beta D_j - cV_j]\Delta t + \sigma_4 V_j\sqrt{\Delta t}\varepsilon_{4,j} + \dfrac{\sigma_4^2}{2}V_j\big(\varepsilon_{4,j}^2 - 1\big)\Delta t, \end{cases} \tag{6.1}$$

where the time increment $\Delta t > 0$, $\varepsilon_{i,j}$ $(i = 1, 2, 3, 4)$ are independent Gaussian random variables which follow the distribution $\mathbb{N}(0, 1)$ for $j = 1, \ldots, n$. We choose realistic parameter values from the published references, and all the parameter values can be seen in Table 2.

Next, by numerical simulations, we mainly pay attention to validate three aspects:
(i) there is a unique ergodic stationary distribution if the condition $R_0^S > 1$ holds;
(ii) the existence of the probability density;
(iii) the effect of large white noise on the number of infected hepatocytes.

Example 6.1 In order to obtain the existence of an ergodic stationary distribution numerically, we choose $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2 = 0.001$ and the other parameter

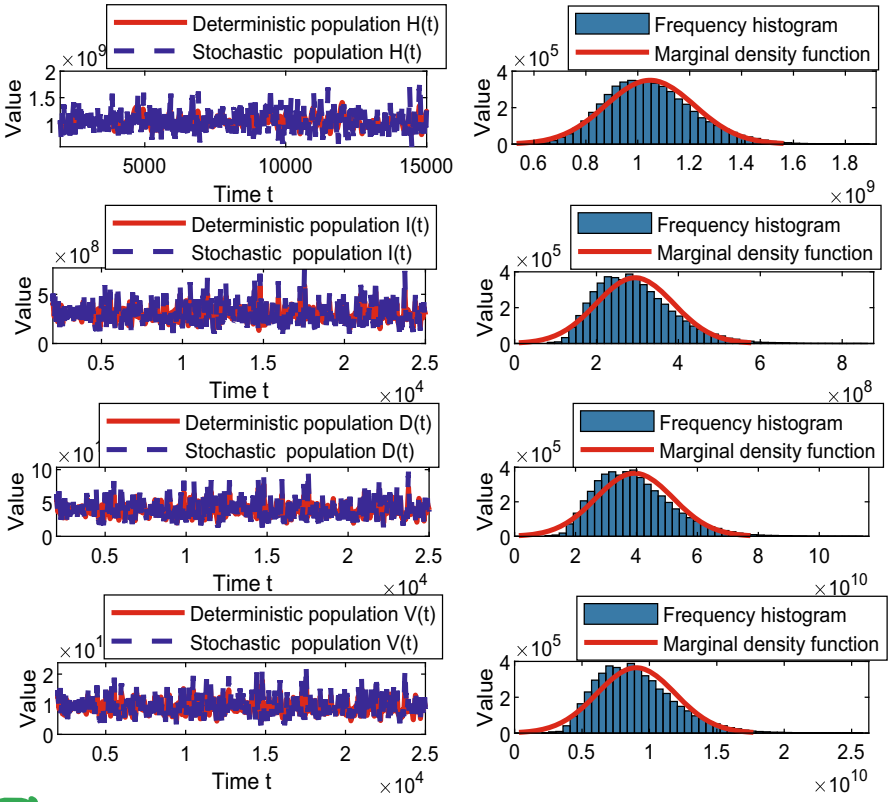

Fig. 1 The left column shows the trajectory of the uninfected hepatocytes, the infected hepatocytes, DNA-containing capsids and DNA-containing virions populations for stochastic model (1.2) and its corresponding deterministic model (1.1) with the noise intensities $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2 = 0.001$. The right column displays the marginal density functions and frequency histograms for each population. Number simulations for the frequency histogram fitting density curves of H, I, D and V of system (1.2) with 10,000,000 iteration points, and the run time of our code is about 14.977289 s

values are shown in Table 2. By direct calculation, we have

$$R_0^S = \frac{a\beta sk}{\left(\mu + \frac{\sigma_1^2}{2}\right)\left(\delta + \frac{\sigma_2^2}{2}\right)\left(\alpha + \beta + \frac{\sigma_3^2}{2}\right)\left(c + \frac{\sigma_4^2}{2}\right)} \approx 2.3687 > 1.$$

That is to say, the conditions of Theorem 3.1 hold. According to Theorem 3.1, we obtain that system (1.2) has a unique ergodic stationary distribution $\pi(\cdot)$ which implies that the disease is persistent a.s. Fig. 1 illustrates this.

Example 6.2 To show the existence of the probability density around the quasi-infected steady state, we choose $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \sigma_4^2 = 0.001$ and the other parameter values are given in Table 2. Direct calculation leads to that $E^* =$ (1,045,392,540.7805, 280,810,809.7534, 37,591,808,534.5861, 8,605,413,346.

9517)^{T} and

$$R_0^S = \frac{a\beta sk}{\left(\mu + \frac{\sigma_1^2}{2}\right)\left(\delta + \frac{\sigma_2^2}{2}\right)\left(\alpha + \beta + \frac{\sigma_3^2}{2}\right)\left(c + \frac{\sigma_4^2}{2}\right)} \approx 2.3687 > 1.$$

In other words, the conditions of Theorem 4.1 are satisfied. Thus, system (1.2) has a log-normal probability density around the quasi-infected steady state E^*. Furthermore, we get $m_1 = 0.1601 \neq 0$, $m_2 = -0.0116 \neq 0$ and $m_3 = 107.6792 \neq 0$. Hence, in view of case (5) of Theorem 4.1, we calculate the specific expression of the covariance matrix Σ,

$$\begin{aligned}\Sigma &= \varrho_1^2 J_1^{-1}\Sigma_0(J_1^{-1})^{\mathrm{T}} + \varrho_2^2(J_3J_2)^{-1}\Sigma_0[(J_3J_2)^{-1}]^{\mathrm{T}}\\ &\quad + \varrho_3^2(J_6J_5J_4)^{-1}\Sigma_0[(J_6J_5J_4)^{-1}]^{\mathrm{T}}\\ &\quad + \varrho_4^2(Q_4Q_3Q_2Q_1)^{-1}\Sigma_0[(Q_4Q_3Q_2Q_1)^{-1}]^{\mathrm{T}}\\ &= \begin{pmatrix} 0.0260 & -0.0088 & -0.0099 & -0.0102\\ -0.0088 & 0.1071 & 0.1066 & 0.1065\\ -0.0099 & 0.1066 & 0.1071 & 0.1069\\ -0.0102 & 0.1065 & 0.1069 & 0.1071\end{pmatrix},\end{aligned}$$

and the corresponding probability density $\Phi(H, I, D, V)$ is as follows:

$$\begin{aligned}&\Phi(H, I, D, V)\\ &= (2\pi)^{-2}|\Sigma|^{-\frac{1}{2}}e^{-\frac{1}{2}(\ln\frac{H}{H^*}, \ln\frac{I}{I^*},}\\ &\ln\frac{D}{D^*}, \ln\frac{V}{V^*})\Sigma^{-1}(\ln\frac{H}{H^*}, \ln\frac{I}{I^*}, \ln\frac{D}{D^*}, \ln\frac{V}{V^*})^{\mathrm{T}}\\ &= 1039.5676\exp\left\{-\frac{1}{2}\left(\ln\frac{H}{1{,}045{,}392{,}540.7805}, \ln\frac{I}{280{,}810{,}809.7534},\right.\right.\\ &\left.\ln\frac{D}{37{,}591{,}808{,}534.5861}, \ln\frac{V}{8{,}605{,}413{,}346.9517}\right)\\ &\times\Sigma^{-1}\left(\ln\frac{H}{1{,}045{,}392{,}540.7805}, \ln\frac{I}{280{,}810{,}809.7534},\right.\\ &\left.\left.\ln\frac{D}{37{,}591{,}808{,}534.5861}, \ln\frac{V}{8{,}605{,}413{,}346.9517}\right)^{\mathrm{T}}\right\}.\end{aligned}$$

See Fig. 2.

Example 6.3 To illustrate that large white noise can lead to the extinction of the infected hepatocytes, we choose $\sigma_1^2 = 0.001$, $\sigma_2^2 = \sigma_3^2 = \sigma_4^2 = 16$ and the other parameter

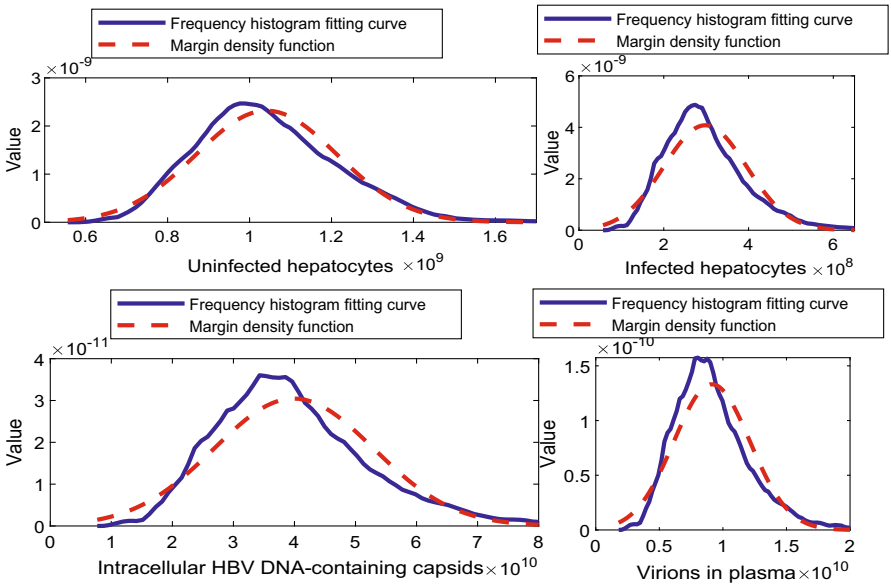

Fig. 2 Numerical simulations for: (i) the frequency histogram fitting density curves of H, I, D and V of system (1.2) with 60,000,000 iteration points, respectively. (ii) The marginal probability densities of H, I, D and V of system (1.2). All of the parameter values are the same as in Fig. 1

values are shown in Table 2. Then, we compute that $\mu = 0.01 > 0.0005 = \sigma_1^2/2$ and

$$\begin{aligned}\nu = &\min\{\delta, \alpha+\beta, c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0\le 1\}} + \max\{\delta, \alpha+\beta, c\}\big(\sqrt[3]{R_0}-1\big)\mathbf{I}_{\{R_0>1\}}\\ &-\big(2\big(\sigma_2^{-2}+\sigma_3^{-2}+\sigma_4^{-2}\big)\big)^{-1}\\ &+\frac{a\beta ks\sigma_1}{\mu\delta R_0^{\frac{2}{3}}(\alpha+\beta)\sqrt{2\mu-\sigma_1^2}} \approx -0.1159 < 0.\end{aligned}$$

Thus, the conditions of Theorem 5.1 hold. By Theorem 5.1, we derive that the infected hepatocytes will go to extinction exponentially with probability one. In addition, by direct calculation, we obtain

$$R_0 = \frac{a\beta sk}{c\delta\mu(\alpha+\beta)} \approx 2.5120 > 1,$$

which implies that all the components of the deterministic system (1.1) will be persistent in a long term. Figure 3 shows this.

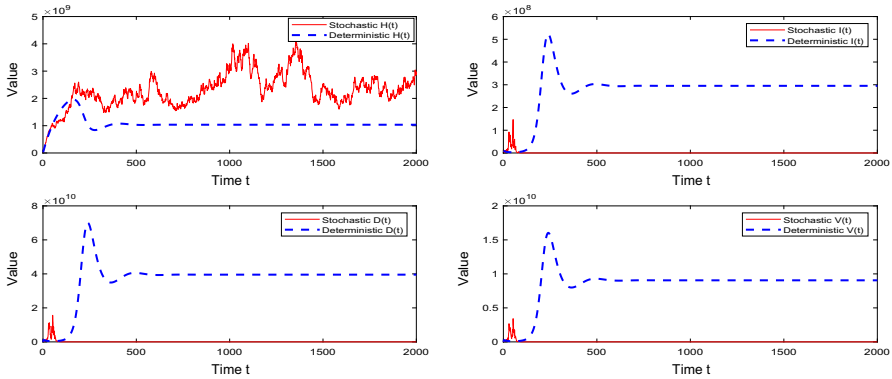

Fig. 3 The solutions of the deterministic system (1.1) and the stochastic system (1.2) with the initial value $(H(0), I(0), D(0), V(0))^{\mathrm{T}} = (0.5, 0.6, 0.4, 0.5)^{\mathrm{T}}$ under the noise intensities $\sigma_1^2 = 0.001$, $\sigma_2^2 = \sigma_3^2 = \sigma_4^2 = 16$

7 Conclusion

In this paper, we develop and study a stochastic HBV infection model with DNA-containing capsids and virions which provides more realistic modeling and analysis approach for the investigation of HBV model. First of all, we prove that system (1.2) has a unique global positive solution with any given positive initial value. Then, we use a stochastic Lyapunov function method to obtain sufficient criteria for the existence and uniqueness of an ergodic stationary distribution, which is a probability distribution with some invariant properties. Especially, under the same conditions as the existence of a stationary distribution, we get the exact expression of the probability density, which is a function that describes the probability of the output value of the random variable around the quasi-infected steady state of system (1.2). Moreover, we derive sufficient conditions for extinction of the infected hepatocytes, which provides the theoretical basis for disease control.

Numerically, based on the actual parameter values in the existing literature, we obtain three important results: (i) small environmental noise makes each population fluctuate very little and so it can retain some stochastic weak stability; (ii) we obtain the accurate expression of the probability density around the quasi-infected steady state of system (1.2); (iii) large white noise can lead to the extinction of the infected hepatocytes which can provide us some new insights for disease control through mathematical theoretical study.

However, there are still many important topics worthy of further study. For example, in this paper, we only obtain the expression of local probability density for system (1.2), but how to obtain the expression of global probability density for system (1.2)? This will attract our great attention. In addition, it is also interesting to study the influence of other types of random perturbations (such as nonlinear perturbations or Poisson jump) on HBV infection model. As far as we know, there is little literature considering four-dimensional HBV infection model with nonlinear perturbations because it is difficult to solve the corresponding Fokker–Planck equation due to the limitations of

mathematical methods. We expect to completely solve it in the future. The relevant work is now underway.

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Availability of Data and Materials Data sharing is not applicable to this article as no datasets are generated or analyzed during the current study.

Declarations

Conflict of interest The author declares that there is no conflict of interest.

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